



Cairo University
Faculty of Computers and Artificial Intelligence

Graduation Project Report

TouRA

By

Alaa Wael Mohamed 20221025

Yussuf Mohammad Taha 20220406

Omar Osama Hassan 20220221

Anas Abdelnasser Ibrahim 20220071

Mohammad Rabea 20220424

Hager Hassan 20220371

Under the Supervisors of:

Dr. Ibrahim Gomaa

Dr. Asmaa Ahmed

TA. Nehal Akram

Department of Information Systems, Faculty of Computers and Artificial Intelligence, Cairo University, Egypt

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This project represents the culmination of teamwork, innovation, and a shared passion for leveraging technology to showcase Egypt's magnificent heritage to the world. We are grateful to everyone who has been part of this transformative journey.

Abstract

Egypt stands as one of the world's most historically and culturally rich destinations, yet tourists continue to face significant challenges that diminish their experience—from poor trip planning and transportation scams to language barriers and limited engagement at historical sites. TOURA is an AI-powered smart tourism platform designed to revolutionize the tourist experience in Egypt by providing a comprehensive, intelligent, and integrated solution that accompanies visitors throughout their entire journey. Unlike traditional tourism platforms that offer isolated services, TOURA leverages advanced artificial intelligence, computer vision, and natural language processing to deliver a seamless end-to-end tourism experience. The platform features an AI-powered trip planner that generates personalized day-by-day itineraries, a trusted booking system for hotels and restaurants, an Uber-like guide service for safe and transparent transportation, and an innovative AR-powered museum companion that brings ancient artifacts to life through image recognition and interactive storytelling. TOURA creates an intelligent conversational agent capable of understanding tourist needs, answering curiosity-driven questions, and providing real-time cultural insights. The computer vision module employs deep learning models trained on Egyptian artifacts to identify monuments and provide rich historical context through simple smartphone photography. While primarily targeting international tourists, TOURA also serves Egyptian citizens and domestic travelers, making cultural heritage accessible to all. The platform's multilingual support breaks down language barriers, while its AI-driven personalization ensures each user receives recommendations tailored to their interests, budget, and travel style.

TOURA represents a paradigm shift from traditional tourism to smart tourism—transforming silent museum visits into interactive learning experiences, replacing tourism chaos with organized guidance, and turning information overload into personalized clarity. Through rigorous development, testing, and iterative refinement, this graduation project aims to enhance Egypt's global tourism image, improve tourist satisfaction, and demonstrate the transformative potential of AI in cultural heritage preservation and tourism innovation.

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1 Introduction

Planning leisure activities and trips has become increasingly dependent on digital platforms. However, most existing solutions focus on transportation and accommodation while overlooking the experience itself, particularly for local exploration and themed outings. Users often struggle to organize meaningful plans that match their mood, social context, or time constraints, and are forced to rely on multiple disconnected applications. This project introduces an intelligent trip planning and guidance system that aims to unify planning and exploration into a single interactive platform.

1.1 Background

With the rapid growth of digital services and smart mobile applications, users increasingly rely on technology to plan travel and leisure activities. While many existing systems focus on traditional travel services such as hotel booking and transportation, they often fail to address the needs of users seeking personalized local experiences, theme-based activity planning, and real-time guidance. Local exploration is often fragmented across multiple platforms for planning, navigation, reviews, and guide services, which leads to a disconnected and inefficient user experience. In addition, many current solutions lack interactive and educational support for visitors at historical and cultural locations. This is especially significant in countries rich in heritage, such as Egypt, where visitors may not always have access to professional guides or immediate information about monuments and historical sites.

1.2 Motivation

1.2.1 Egypt's Untapped Tourism Potential

Egypt is among the world's richest countries in historical, cultural, and natural tourism resources, hosting iconic landmarks such as the Pyramids of Giza, the temples of Luxor, and the Red Sea resorts. Despite these exceptional assets, Egypt's tourism performance remains below its true potential. International comparisons show that tourist satisfaction and economic returns depend not only on attractions, but primarily on the quality of the overall tourism experience. Modernizing tourism services through intelligent digital platforms can significantly enhance visitor satisfaction, increase tourist spending, improve Egypt's global tourism ranking, and support sustainable tourism development.

1.2.2 Opportunity Created by Technological Advancement

Recent advances in artificial intelligence, computer vision and natural language processing have created new possibilities for transforming tourism services. These technologies enable intelligent trip planning, interactive cultural exploration, and real-time personalized assistance

through smartphones. The convergence of these technologies makes it feasible to develop an integrated smart tourism companion, positioning Egypt as a leader in digital tourism innovation.

1.2.3 Addressing a Critical Market Gap

No existing platform provides a comprehensive, AI-powered, end-to-end tourism solution that addresses all major tourist pain points in a single integrated experience. TOURA fills this critical market gap by combining trip planning, trusted booking, transportation, and interactive cultural exploration in one intelligent mobile application.

1.2.4 Cultural Heritage Preservation

TOURA contributes to cultural preservation by transforming passive museum visits into interactive learning experiences. By improving access to historical knowledge, the platform enhances global understanding of Egyptian civilization and promotes respectful, informed tourism. Technology thus becomes a tool for preserving and promoting Egypt's cultural identity in the digital age.

1.2.5 Economic and Social Impact

Smart tourism platforms have demonstrated strong economic benefits, including increased tourist arrivals, higher spending per visitor, job creation, and stronger support for local businesses. By improving service quality and accessibility, TOURA contributes directly to national economic development and sustainable tourism growth in Egypt.

1.3 Problem Statement

Despite the availability of numerous travel and navigation applications, there is no comprehensive solution that effectively integrates theme-based trip generation, local activity planning, guide coordination, and AI-assisted exploration into a single system. Current systems suffer from several limitations, including the lack of personalized plans based on user preferences, insufficient support for local and short-term trips, limited interaction between users and professional guides, and minimal real-time visualization of trip progress. Moreover, tourists visiting historical locations often lack instant access to accurate and contextual information.

These challenges create a gap between user expectations and the capabilities of existing systems. Therefore, there is a need for an intelligent trip planning application that bridges this

gap by offering personalized planning, real-time guidance, and educational support through advanced technologies.

1.4 Objectives

The primary objective of this project is to design and implement an intelligent trip planning and guidance system that enhances both local and travel experiences.

The specific objectives of the project include:

- Providing users with the ability to create trip plans based on selected themes such as romantic, family, friends, or solo.
- Generating intelligent plans that distribute activities according to the chosen theme.
- Allowing users to select locations using geographical coordinates, cities, or government regions.
- Enabling professional guides to bid on user-created plans through a bidding system.
- Supporting mutual rating and feedback between users and guides.
- Displaying trip routes, visited locations, and guide positions on an interactive map.
- Allowing users to rate and review places included in their plans.
- Integrating artificial intelligence to recognize monuments and statues from images and provide historical information.
- Supporting AI-assisted chat interaction for requesting and retrieving trip plans from the system.

1.5 Project Scope and Limitations

1.5.1 Project Scope

The scope of this project includes the development of a multi-functional platform that supports users, guides, and place owners. The system covers trip creation, theme-based plan generation, guide bidding, map visualization, ratings and reviews, AI-based image recognition, and conversational interaction with the system.

1.5.2 Project Limitations

1.5.2.1 Data Collection and Quality

The success of the proposed smart tourism platform fundamentally depends on the availability of large, accurate, and trustworthy datasets. However, one of the most critical obstacles in the Egyptian tourism sector is the severe lack of structured, reliable data. This problem affects two core components of the system: historical artifact recognition and trusted tourism services.

Lack of Structured Datasets for Historical Artifacts A major technical challenge is the absence of a comprehensive digital dataset of Egyptian statues, monuments, and museum artifacts. Although Egypt possesses thousands of invaluable historical objects, most of them are not

digitally documented in a form suitable for storing and using in our model. There is no centralized, open, and standardized dataset that contains high-quality images of artifacts with verified historical labels. Existing images are scattered across the internet, often poorly labeled, inconsistent, or historically inaccurate.

As a result, building an artifact recognition system is not merely a modeling task, but a data creation problem. It requires constructing a dataset from scratch by systematically collecting images from museums and historical sites and ensuring expert-level historical verification. The absence of ready-to-use datasets significantly increases the technical complexity of the project and represents a major barrier to accurate AI deployment. Absence of a Trusted Digital Database of Tourism Services An equally serious challenge is the lack of a reliable and verified digital database of hotels, restaurants, tour guides, and tour operators in Egypt. Unlike developed tourism markets, Egypt does not possess a centralized, trustworthy repository of tourism service providers. Most available online information is fragmented, outdated, or biased. User-generated reviews alone are insufficient, as they are vulnerable to manipulation, fake ratings, and incomplete coverage. Consequently, there is no dependable data source that can be directly used to build an AI-driven recommendation and booking system. This means that the platform cannot simply “use existing data,” but must create a trusted database from the ground up. This involves verifying service legitimacy, pricing fairness, and safety standards—tasks that cannot be fully automated and require structured validation processes. The absence of such a foundation makes data acquisition one of the most difficult and resource-intensive aspects of the project.

1.5.2.2 Technical Challenges

LLM Application Development

Developing a reliable, production-grade LLM-based system using LangChain and LangGraph presents several technical challenges. These include effective context management to preserve conversation history and user preferences across interactions, and hallucination control to ensure the accuracy of historical and cultural information. Additionally, the system must balance response quality with low latency to meet real-time mobile application requirements. Integrating multiple AI tools such as trip planning, question answering, and service recommendations within a unified conversational framework further increases system

complexity. Finally, careful prompt engineering is required to ensure that responses are relevant, culturally appropriate, and factually reliable.

1.6 Project Methodology

In this project, the **Agile methodology** was adopted due to its iterative and flexible nature. Agile supports incremental development, allowing system components to be designed, implemented, and tested in short cycles while accommodating evolving requirements. This approach is particularly suitable for the proposed system, which includes interactive and intelligent features that benefit from continuous feedback and refinement. Additionally, Agile promotes early testing and continuous integration, improving system quality and reducing development risks.

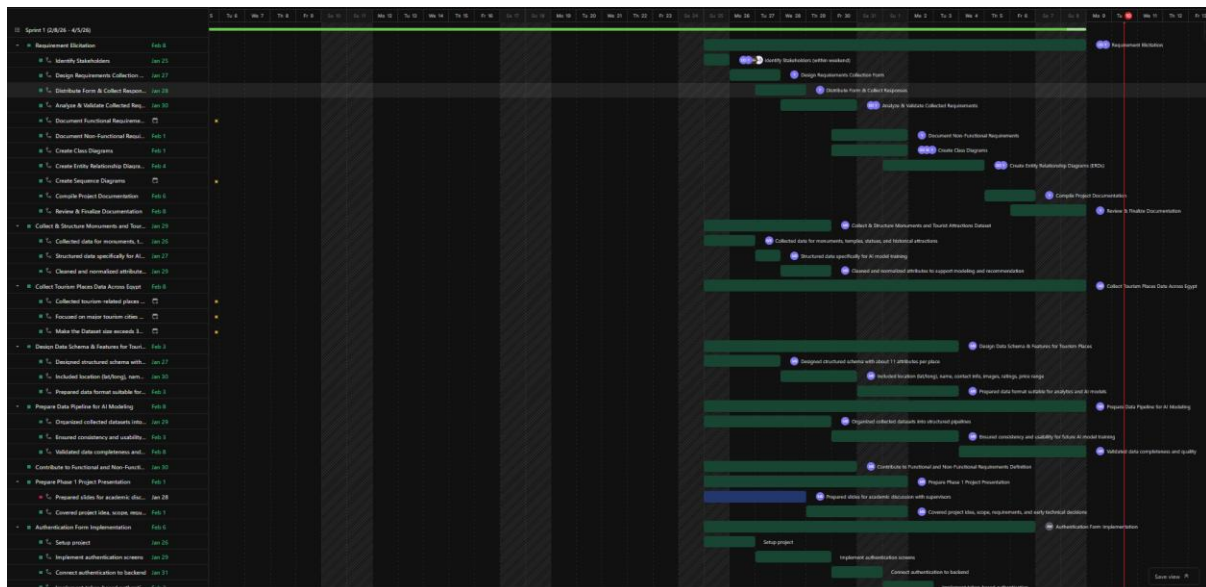


Figure 1-1 Gantt chart for milestones

1.7 Project Report Outline

- Chapter 2 presents the literature review and Market analysis of TouRA
- Chapter 3 presents the analysis of Project stakeholders, use case diagrams, uml class diagrams and ERDS
- Chapter 4 presents the AI methodology used in the project
- Chapter 5 presents UI UX for the phone application

2 Market and Literature Survey

The development of TOURA builds upon existing research and commercial applications in several intersecting domains: AI-powered travel planning, computer vision for cultural heritage, conversational AI systems, and mobile tourism platforms. This literature review examines the current state of the art in these areas, identifies gaps in existing solutions, and establishes how TOURA advances beyond current offerings to provide a comprehensive smart tourism ecosystem.

2.1 Related Works

1. Wanderlog

represents the current generation of digital trip planning tools. The platform enables tourists to generate multi-day itineraries for Egypt and other destinations, incorporating recommendations for hotels, restaurants, and activities. Wanderlog's strength lies in its ability to consolidate the booking process, allowing users to handle their entire trip through a single platform interface.

However, Wanderlog and similar platforms suffer from several critical limitations:

- **Generic recommendations:** Itineraries are generated based on general popularity data rather than personalized AI-driven analysis of individual user preferences, travel styles, and interests
- **Limited local context:** The platform lacks deep integration with local Egyptian tourism infrastructure and cultural insights
- **No on-ground support:** Once tourists arrive at their destination, these platforms provide minimal real-time guidance or assistance

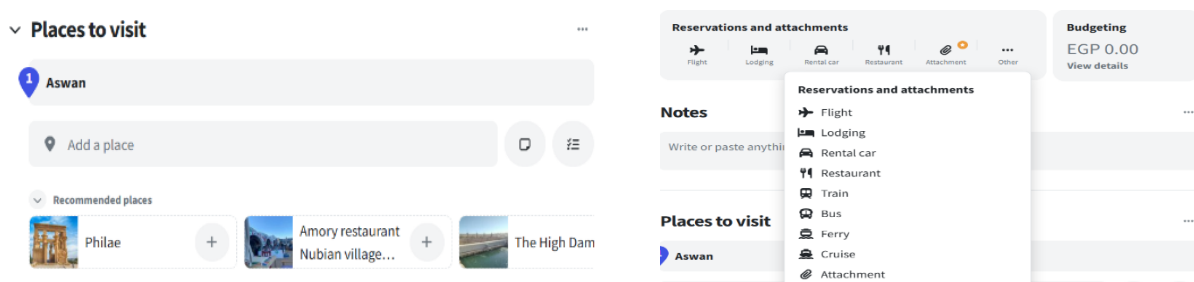


Figure 2-1 Wanderlog

2. Egymonuments

takes a narrower approach by focusing specifically on online ticketing for Egyptian monuments and archaeological sites. The platform's website invites users to "book your monument(s) ticket(s)" in advance, addressing a specific friction point in the tourism experience. While useful for reducing queue times and ensuring entry to popular sites, Egymonuments remains a single-purpose utility rather than a comprehensive tourism solution. It does not address trip planning, transportation, accommodation, cultural interpretation, or any of the broader challenges tourists face.

3. Tripadvisor

stands as one of the most established players in the tourism technology space.

The platform combines user-generated reviews with booking capabilities, helping tourists generate multi-day Egypt itineraries with recommended hotels, restaurants, and activities. TripAdvisor's primary strength is its extensive review database, which provides social proof and helps tourists make informed decisions.

Critical analysis reveals several limitations:

- **Review reliability issues:** The platform struggles with fake reviews, incentivized reviews, and reviews from users with vastly different expectations than the current tourist
- **Information overload:** Users face thousands of reviews and options without intelligent filtering or personalization
- **Reactive rather than proactive:** The platform responds to user searches but does not proactively guide trip planning based on comprehensive understanding of user preferences
- **Limited Egypt-specific optimization:** While covering Egypt, the platform does not specialize in Egyptian tourism challenges such as scam prevention, cultural context, or Arabic-English language bridging

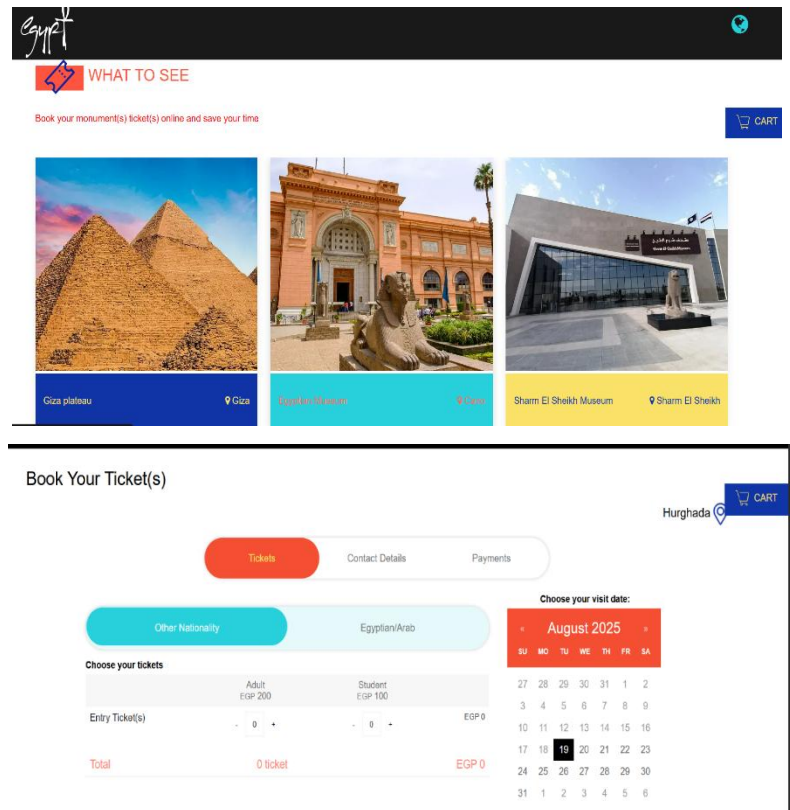


Figure 2-2 Egymonuments

- **No interactive museum experience:** TripAdvisor provides reviews of museums but offers nothing to enhance the actual visit experience

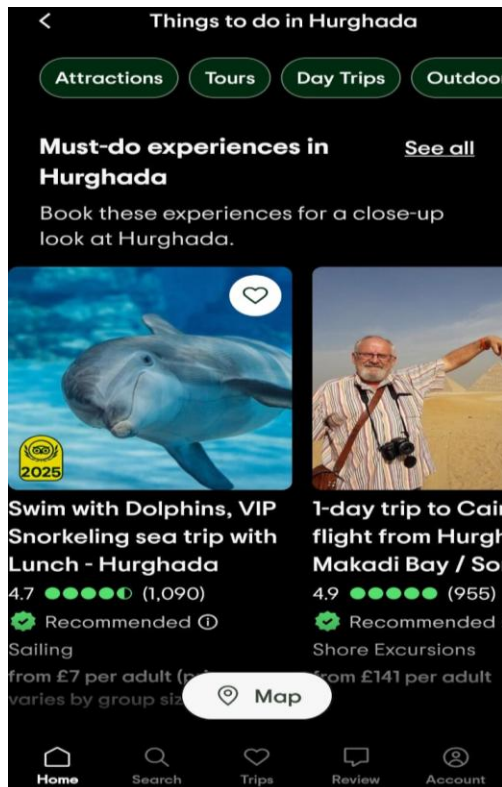


Figure 2-3 Tripadvisor

4. Egypt Tour Guide

represents an Egypt-specific mobile application containing travel maps, attraction lists, and information about top tourist sites. The app categorizes attractions by type (Park, Nature, Museum, Religion, Family, Architecture, Sculpture, Viewpoint) and provides several useful features:

Features include:

- Viewing top attractions on maps with brief introductions
- Listing attraction sites by distance from the map center
- Filtering attractions by city, category, or keywords
- Adding attractions to favorites
- "What To Do" lists
- Offline content data

While Egypt Tour Guide addresses some Egypt-specific needs, it remains fundamentally a digital guidebook rather than an intelligent assistant. The application provides static information without personalization, does not facilitate bookings or transportation, offers only "brief information" rather than deep cultural context, and lacks any AI-powered interactivity or visual recognition capabilities. The offline capability is valuable, but the content itself is predetermined rather than dynamically generated based on user needs.

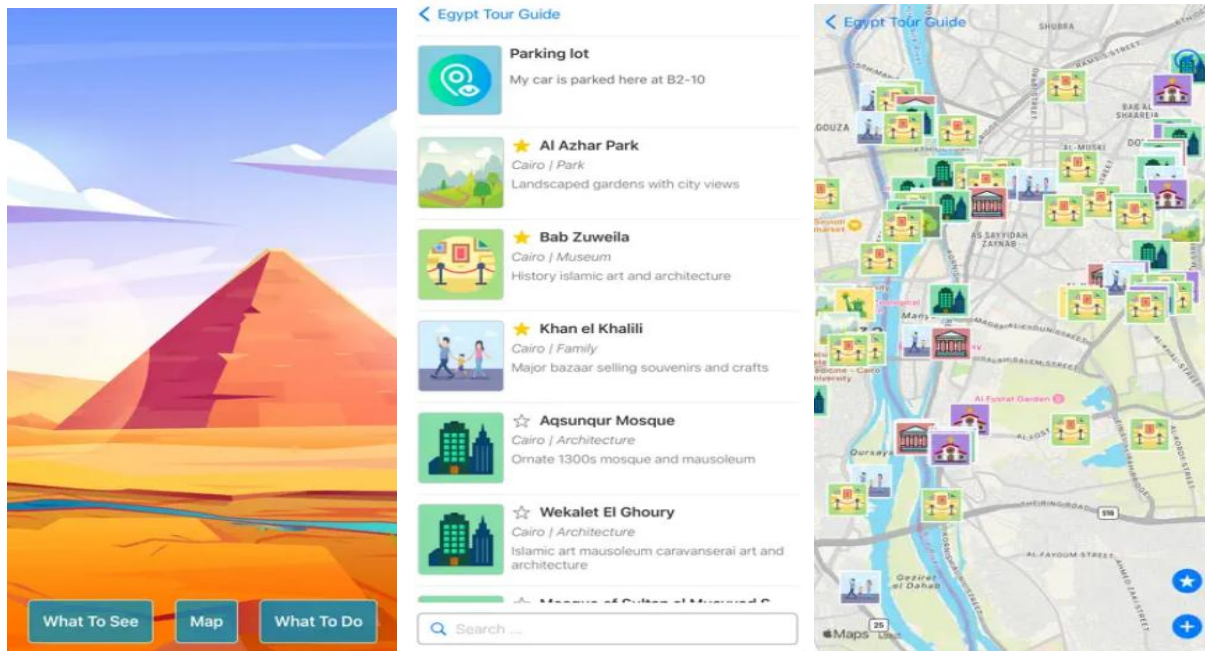


Figure 2-4 Egypt Tour Guide

5. AI Trip & Itinerary Planner

represents a more advanced approach to travel planning, incorporating artificial intelligence to create personalized itineraries. The application markets itself as a "personalized travel assistant" that generates tailor-made itineraries based on user interests, preferences, and location.

Key features include:

- Visual landmark recognition
- Real-time information delivery
- Multilingual support
- Essential travel resources (emergency contacts, checklists)

This application demonstrates important advancements in applying AI to tourism, particularly in personalization and visual recognition. However, several gaps remain:

- **Generic global focus:** The app covers worldwide travel rather than specializing in specific destinations, limiting depth of local knowledge and cultural context
- **Recognition scope:** While offering visual landmark recognition, the system focuses on famous landmarks rather than the comprehensive artifact and monument recognition needed for museum experiences
- **Limited cultural depth:** The app provides practical travel assistance but does not deeply engage with cultural heritage, historical storytelling, or educational content
- **No service provider integration:** The platform does not verify or vet service providers, address scam prevention, or ensure quality standards
- **Missing transportation solution:** No integrated transportation or guide service component



Figure 2-5 AI Trip & Itinerary Planner

6. LandSnap – Identify Landmarks

is an iPhone application that leverages artificial intelligence to recognize famous landmarks and attractions from user-provided images. Users can either photograph landmarks with the device camera or upload existing photos for recognition. The application claims recognition support for over 120,000 global landmarks including iconic structures such as the Statue of Liberty, the Great Pyramid of Giza, and the Sydney Opera House.

Key technical characteristics:

- AI-powered photo-based identification
- Large-scale recognition database (120,000+ landmarks)
- Simple user interaction model
- Subscription-based monetization (monthly, quarterly, yearly)

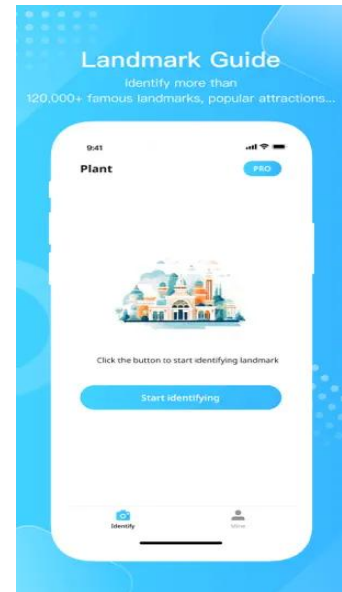


Figure 2-6 LandSnap

Analysis reveals significant limitations:

- **Surface-level identification:** The app identifies landmarks by name but provides minimal historical context, cultural significance, or detailed information
- **Limited engagement:** Recognition is a one-time lookup rather than an interactive learning experience
- **Narrow functionality:** The app focuses exclusively on famous landmarks, not museum artifacts, archaeological details, or less prominent cultural sites
- **Low market adoption:** Limited user engagement and reviews suggest the narrow utility has not achieved significant market traction
- **No integration with broader travel needs:** Landmark identification exists in isolation without connecting to trip planning, booking, or navigation

7. AI Tour – Landmark Identifier

ffers a similar approach with enhanced informational content. After photographing a landmark, the app's AI model identifies it and returns detailed information including the name, history, and facts. AI Tour positions itself as a travel companion that enhances exploration through "captivating stories" and historical context.

Distinguishing features:

- Richer content delivery beyond simple identification
- Focus on storytelling and educational value

2.1.1 Subscription model for full functionality

Despite improvements over LandSnap, AI Tour still exhibits critical gaps:

- **Limited to major landmarks:** Does not extend to museum artifacts, which represent the majority of tourist cultural engagement in Egypt
- **Generic historical content:** Information appears to be pre-written rather than dynamically generated or personalized to user interests
- **No conversation capability:** Users receive static information rather than being able to ask follow-up questions
- **Minimal market presence:** Low adoption suggests the standalone landmark identification value proposition is insufficient

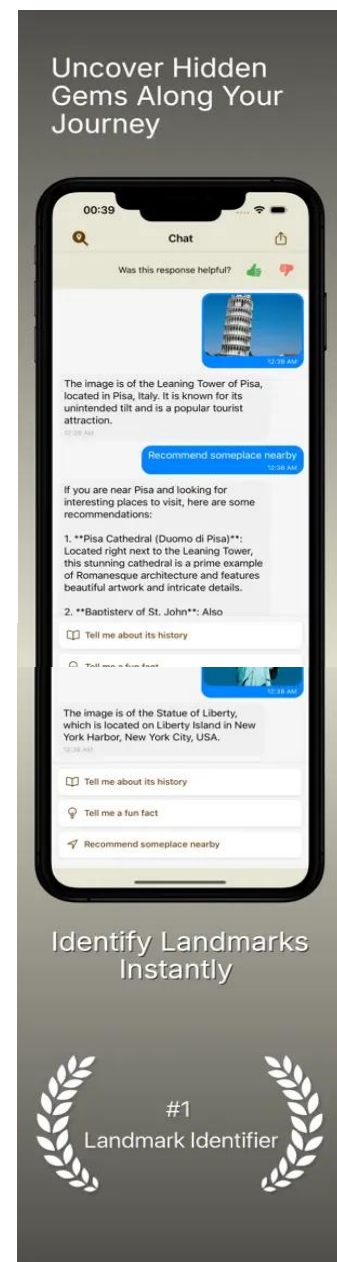


Figure 2-7 AI Tour

2.2 LLMs

1. Gemini

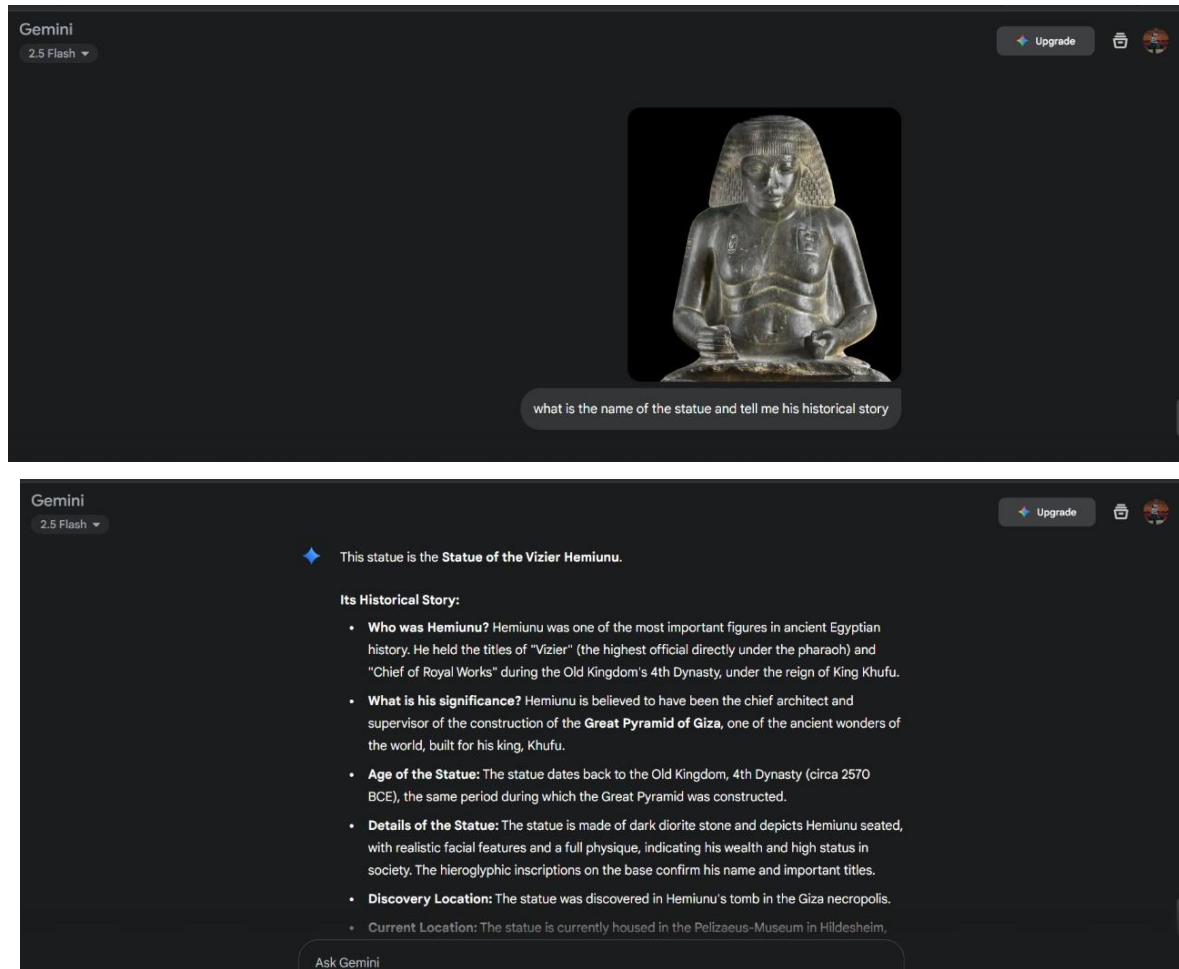


Figure 2-8 Gemini's reply when prompting it with an image

2. ChatGpt

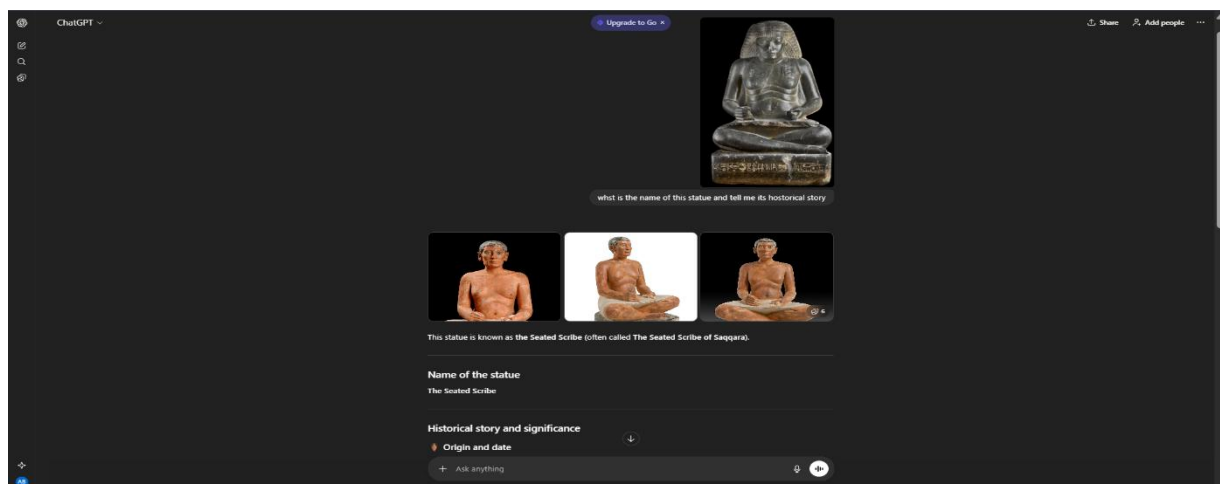


Figure 2-9 Chatgpt's reply when prompting it with an image

2.3 Comparison table between TouRA and related works

Feature	Wanderlog	Egy monuments	TripAdvisor	Egypt Tour Guide	AI Trip & Itinerary Planner	LandSnap	AI Tour	Proposed System
Trip Planning	✓	✗	✓	✓	✓	✗	✗	✓
Theme-Based Planning	✗	✗	✗	✗	✓	✗	✗	✓
Interactive AI Chat	✗	✗	✗	✗	✗	✗	✗	✓
Egypt-Specific Focus	✗	✓	✗	✓	✗	✗	✗	✓
Guide Marketplace	✗	✗	✗	✗	✗	✗	✗	✓
Map-Based Plan Visualization	✓	✗	✓	✓	✓	✗	✗	✓
Reviews Integration	✗	✗	✓	✗	✗	✗	✗	✓
AI Monument Recognition	✗	✗	✗	✗	✓ (Limited)	✓	✓	✓ (Limited)
Deep Cultural / Historical Content	✗	✗	✗	✓ (Limited)	✓ (Limited)	✗	✓	✓

Table 2-1 TouRA vs Related Works

2.4 Market Gaps

Based on the analysis of existing platforms, several important gaps in the current market were identified. While many applications provide partial solutions for trip planning or tourism support, none deliver a complete, intelligent, and locally optimized experience that matches modern user expectations.

Lack of Theme based planning:

Most current systems rely on general popularity or static recommendations rather than adapting to user mood, social context, or preferences. Survey responses showed that many users face difficulty deciding where to go and struggle to find places that match the desired vibe (romantic, family, chill, etc.), highlighting the need for AI-driven, theme-based planning.

Fragmented Planning Process:

Users typically depend on multiple apps for a single outing — one for ideas, another for maps, another for reviews, and sometimes separate services for guides. This fragmentation leads to time loss, confusion, and information overload. The market lacks a unified system that integrates all these services into one platform.

Trust and Review Reliability Issues:

Survey participants frequently mentioned poor or fake reviews as a major issue. Existing platforms do not sufficiently ensure review credibility or provide strong verification mechanisms for guides and service providers, reducing user trust.

Limited On-Ground and Real-Time Support:

Most planning tools focus on pre-trip organization but offer little support once users reach the destination. There is a gap in real-time guidance, interactive map-based plans, and immediate AI assistance during the outing.

Weak Cultural and Educational Interaction:

Landmark recognition apps identify famous places but provide shallow or static information. They do not offer interactive learning, storytelling, or conversation-based AI support, especially for culturally rich environments such as Egyptian historical sites.

Lack of Integrated Guide Marketplace:

While users may want local guides, there is no structured system where guides can offer services competitively with ratings, verification, and transparent selection.

2.5 Survey Analysis

Since we have no specific data on how the app would be effective, we conducted a survey to get the data from it, and we got 30 responses in return.

We found that 93% respondents fall within the 18–24 age group, with a smaller portion between 25–34. Most participants are students and young employees, which represents an active, socially engaged demographic that frequently goes out for leisure activities such as restaurants, cafés, parks, and entertainment places. This group is relevant to the target market of the application.

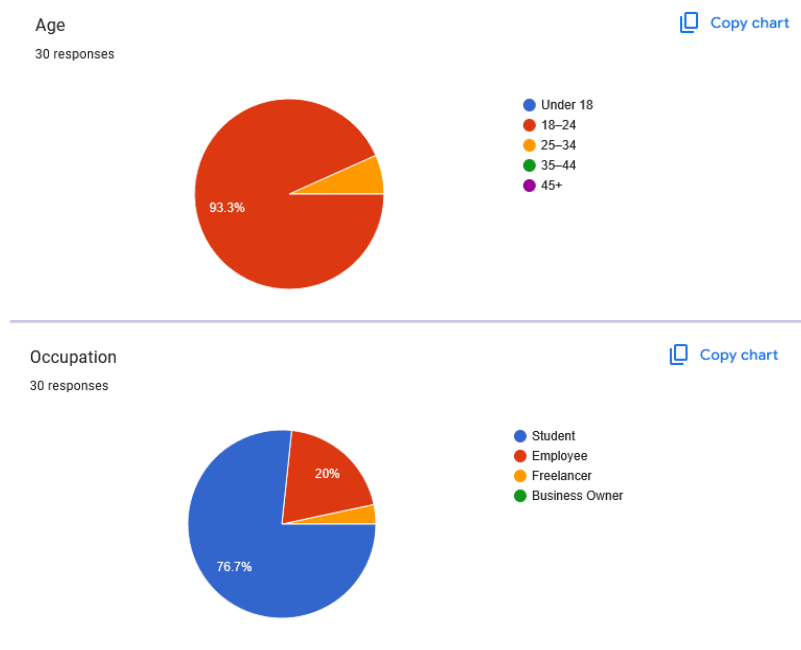


Figure 2-10 Age and Occupation of Respondents

Also, we found that 60% of the respondents use multiple apps to plan an outing

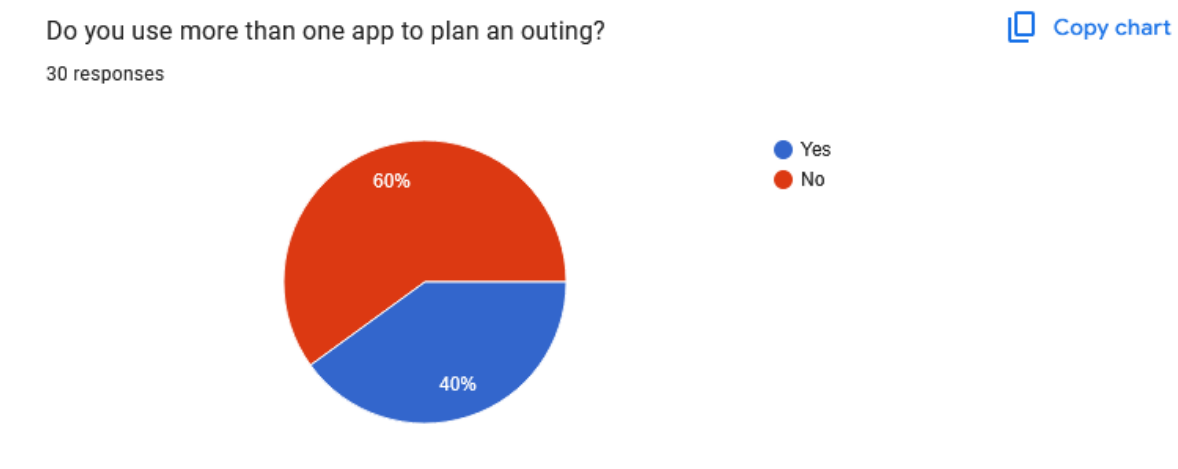


Figure 2-11 How many use more than one app

60% of the respondents show interest in using the app to plan their day based on a theme

Would you use an app that creates a full outing plan for you based on a theme (romantic, family, friends, solo)?

[Copy chart](#)

30 responses

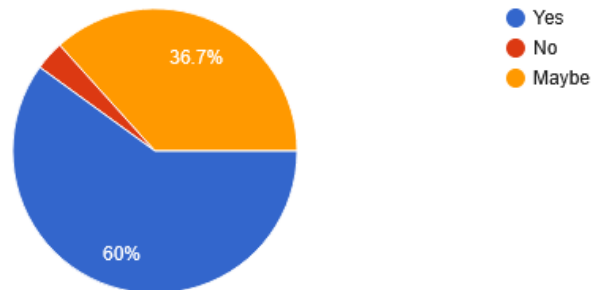


Figure 2-12 How much interest in the app

Based on the survey, the idea of a guide for a trip is not very interesting unless it's a trip related to historical places

Would you use an app where guides can bid to guide you during a trip? (like InDrive)

[Copy chart](#)

30 responses

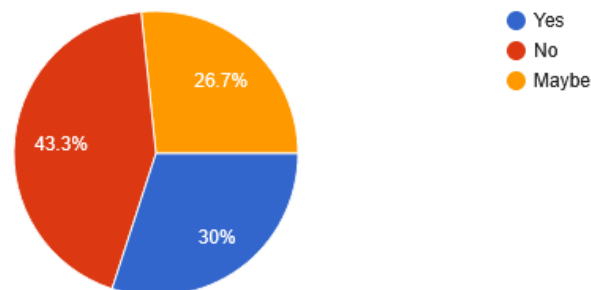


Figure 2-13 Pie chart of bidding a guide

This answer also supports the above graph

Any suggestions for features you would like to see?

maybe keep the guides for historical or monuments, no one wants to see a guide for a casual trip with friends down to the coffee shop and mall for example.

Figure 2-14 Feature Suggestion from respondents

60% of the respondents find that taking a picture of a monument and send it to an AI is interesting

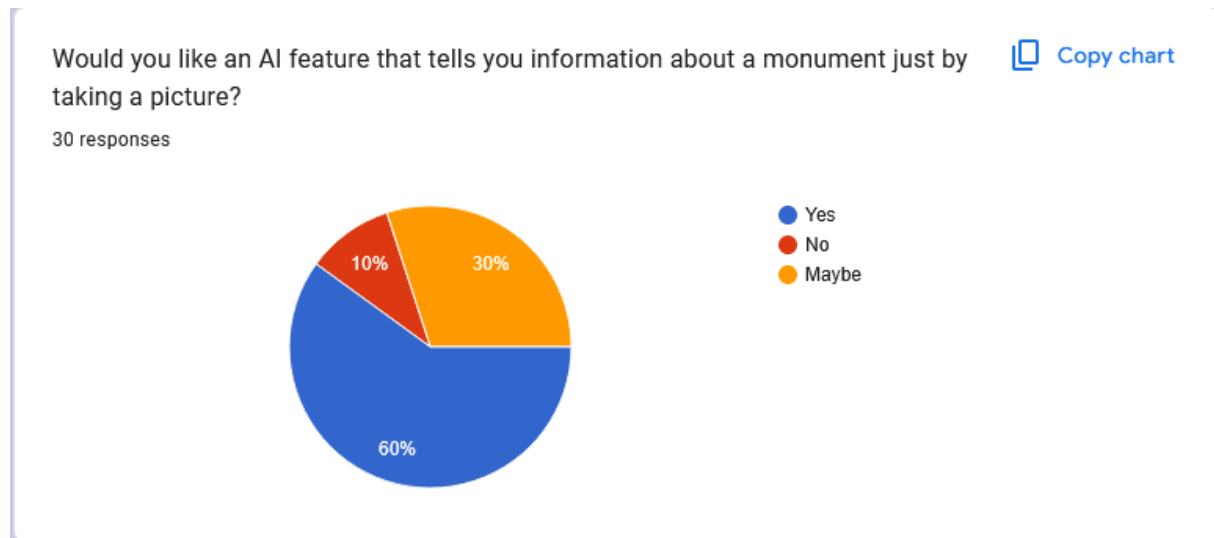


Figure 2-15 Pie Chart of Response to taking a monument picture

Almost 60% of respondents like the idea of asking an AI to generate an outing plan

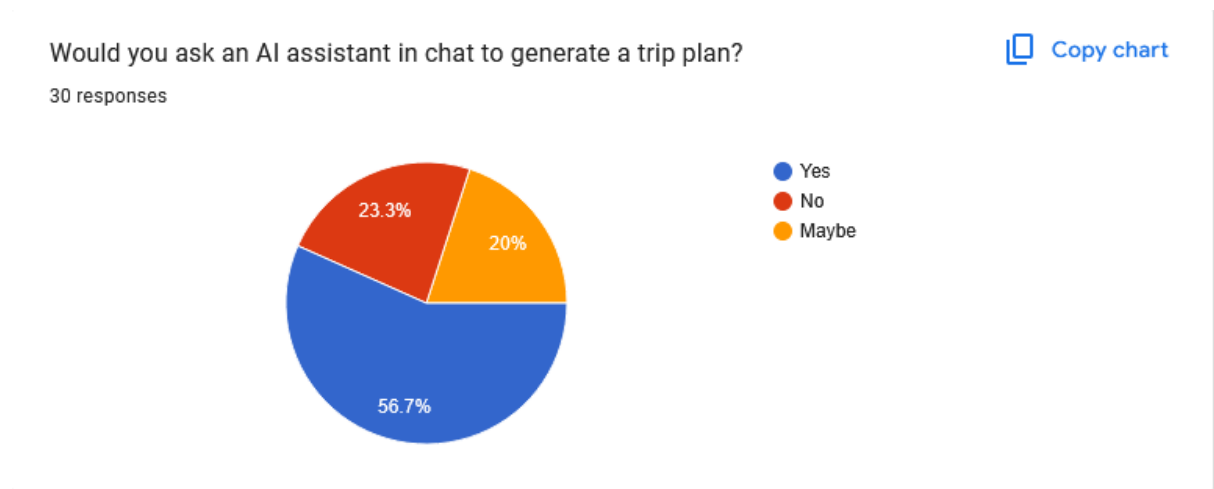


Figure 2-16 How many in favor of chatting with an AI

Most of the respondents think that the app is useful and would download it.

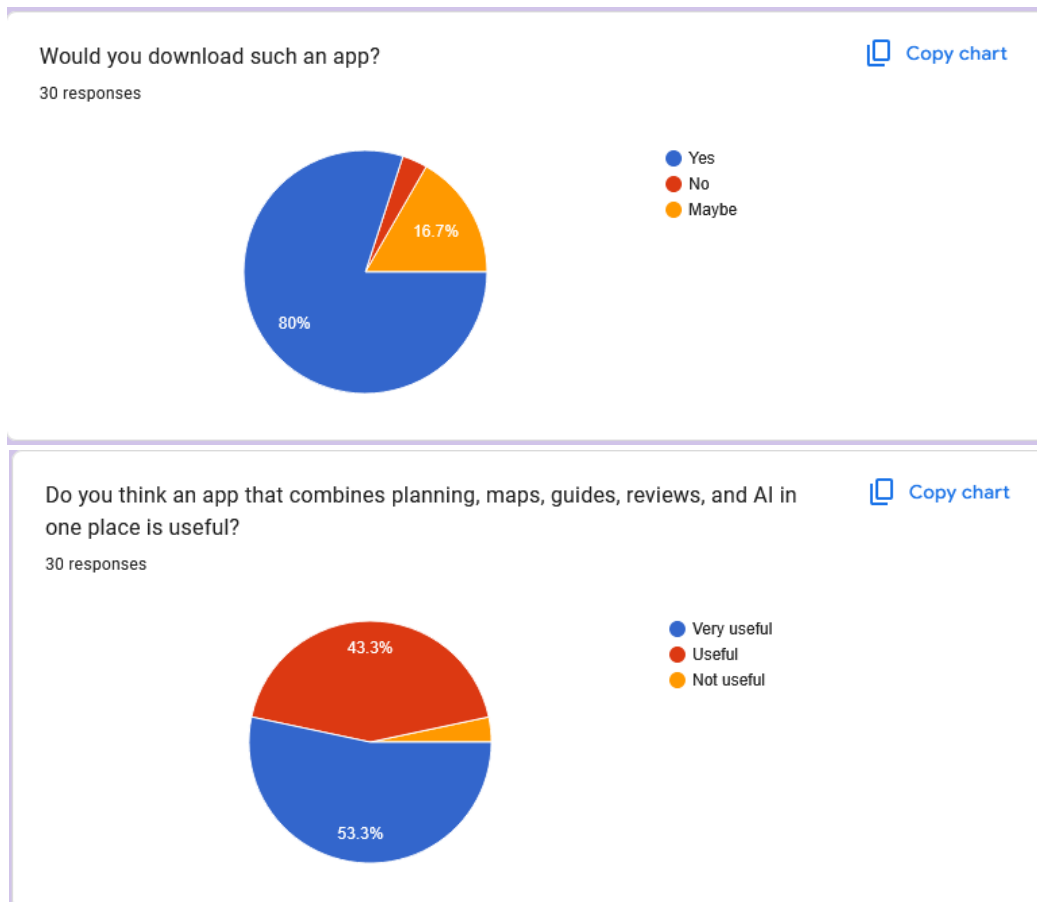


Figure 2-17 How welcomed is the idea

2.6 Project Description (Theoretical Overview)

TouRA is a smart outing and trip planning application designed to simplify how users choose destinations and organize activities. It provides an integrated platform that combines location discovery, trip planning, mapping, user reviews, local guide services, and artificial intelligence features in one system.

The main idea is to move users away from scattered planning methods that rely on multiple apps and sources. Instead, the application generates a complete outing plan based on themes and preferences such as romantic, family, friends, or solo activities. Users enter details like theme, location, and budget, and the system suggests suitable places arranged into an organized plan displayed on an interactive map.

The system also includes a guide feature that connects users with local guides, supported by ratings, reviews, and verification for trust. AI capabilities are integrated through a chat assistant for trip planning and an image-based feature that provides information about landmarks.

Theoretically, the project integrates recommendation systems, location-based services, social feedback, and AI-driven interaction into one unified platform to reduce planning effort and improve the outing experience.

3 TouRA Analysis

This chapter defines the TouRA's technical and functional specifications. It outlines the functional and non-functional requirements, identifies key stakeholders, and presents system scenarios to illustrate real-world interactions. This chapter includes use case diagrams, sequence diagrams, and class diagrams to provide a structured visualization of the system's design and behavior.

3.1 Project Stakeholders

End Users (Travelers): Use the application to plan trips, explore locations, request guides, and use AI features. Their experience and satisfaction are the focus of the system.

Tour Guides: Provide guiding services, bid on trips, and build their reputation through ratings and reviews.

Companies: Organizations listing packages, managing multiple guides, and handling bulk bookings and operations

Place Owners: Manage and update information about their businesses after ownership approval, benefiting from visibility and user engagement.

Administrators: Oversee the platform, manage users and content, approve ownership requests, and ensure proper system operation.

Development Team: Responsible for designing, developing, maintaining, and improving the system while ensuring performance, security, and scalability.

3.2 Functional Requirements

- User Management

- Identity and Access Management

- The system shall use Keycloak as the identity and access management solution.
- The system shall issue JWT access tokens upon successful authentication.
- Tokens shall include user identifier, role, and session metadata.
- Authentication flows shall be handled through a dedicated User Service.

- User Registration with OTP Verification

- The system shall allow users to register using either email or phone number.
- Upon registration request, the system shall generate a one-time password (OTP).
- The Notification Service shall send the OTP via email or SMS depending on the provided method.
- The user shall verify the OTP to complete registration.
- Upon successful OTP verification, the user account shall be activated in Keycloak and a JWT token shall be issued.
- OTPs shall be time-bound and single-use.

- User Login

- The system shall allow users to log in using email or phone and password.
- Login shall not require OTP verification.
- Upon successful login, the system shall issue a JWT token via Keycloak.
- The login response shall include a profile setup flag.
- If profile setup is incomplete, the user shall be redirected to profile setup; otherwise, to the home page.

- **Forgot Password and Password Reset**

- The system shall provide a forgot password flow using OTP verification.
- Users shall request password reset using email or phone number.
- The system shall generate and send an OTP via the Notification Service.
- After OTP verification, the user shall be allowed to log in.
- The user shall be required to change their password after login.
- The system shall invalidate the reset token once the password is updated.

- **Service Communication**

- The User Service shall handle registration, authentication, OTP validation, and token issuance.
- The Notification Service shall handle SMS and email provider integration.

- **Trip Planning**

- **Location Selection**

- The user shall be able to select a city, governorate, or provide latitude and longitude.
 - The system shall detect the user's current location if permission is granted.

- **Plan Themes**

- The user shall select a theme such as romantic, family, friends, or solo.
 - The system shall apply theme rules such as percentages of place types.

- **Plan Generation**

- The system shall generate a trip plan based on theme rules, location, and available places.
 - The system shall ensure the plan contains diverse activities in the correct theme percentages.
 - The user shall be able to regenerate or edit the plan.

- Featured and Custom Plans

- Plan Ownership

- Each plan shall have an owner, either a Company or a Guide.
- Owner information shall be stored and displayed to the user.
- Companies can assign multiple guides to their plans.
- Guides owning plans automatically assign themselves.
- Multiple guides per plan shall be allowed only for company-owned plans.

- Plan Details

- Each plan shall include name, theme, list of places, owner information, and optional assigned guides.
- Admins can mark a plan as a featured plan.

- Guide Assignment

- Companies can assign multiple guides to their plans.
- Guides owning a plan automatically become assigned guides.
- Assigned guides can view upcoming trips associated with the plan.

- User-Generated and AI Plans

- Users can generate trip plans using AI or an algorithm.
- Generated plans shall be owned by the creating user.
- Other users can join published plans until the participant limit is reached.
- The creator can request a guide once enough users join.
- Guides can bid on requested plans.
- A group chat shall be automatically created for participants.
- Plans shall start at the scheduled time or when a guide is assigned.

- Maps and Navigation

- The system shall display all places in a plan on an interactive map.
- Users shall view location details including address, rating, and description.
- If a guide is hired, the system shall display the guide's live location.
- Guides shall be able to enable or disable live location sharing.

- Guide Marketplace

- Users shall be able to request a guide for a plan.
- Guides shall view requests and place bids.
- Users shall accept one bid and reject the others automatically.
- The system shall lock the plan once a guide is assigned.

- Ratings and Reviews

- Users and guides shall rate and review each other after a trip.
- Ratings shall be publicly displayed on profiles.
- Users shall rate places they visited within a plan.
- Ratings shall influence future plan-generation rankings.

- AI Features

- Users shall upload images of monuments or statues.
- The system shall identify monuments and return historical and cultural information.
- Users shall request trip plans via chat.
- The system shall generate and display trip plans within the chat interface.

3.3 Non-Functional Requirements

- Performance

- The system should support at least 2,000 concurrent users.
- Map rendering should complete within two seconds.
- AI responses should be returned within five to eight seconds.

- Security

- All data transmission shall be encrypted using HTTPS and TLS.
- Only authenticated users shall access protected endpoints.
- Administrative endpoints shall require role-based authorization.

- Reliability

- The system shall maintain an uptime of at least 99.5%.
- Real-time features shall recover automatically after disconnection.

- Usability

- The map interface shall be mobile-friendly.
- Trip creation shall not exceed four steps.

- Maintainability

- The system shall follow Clean Architecture and CQRS principles.

- AI Quality

- Monument recognition accuracy shall exceed 85%.

3.4 Use Case Diagram

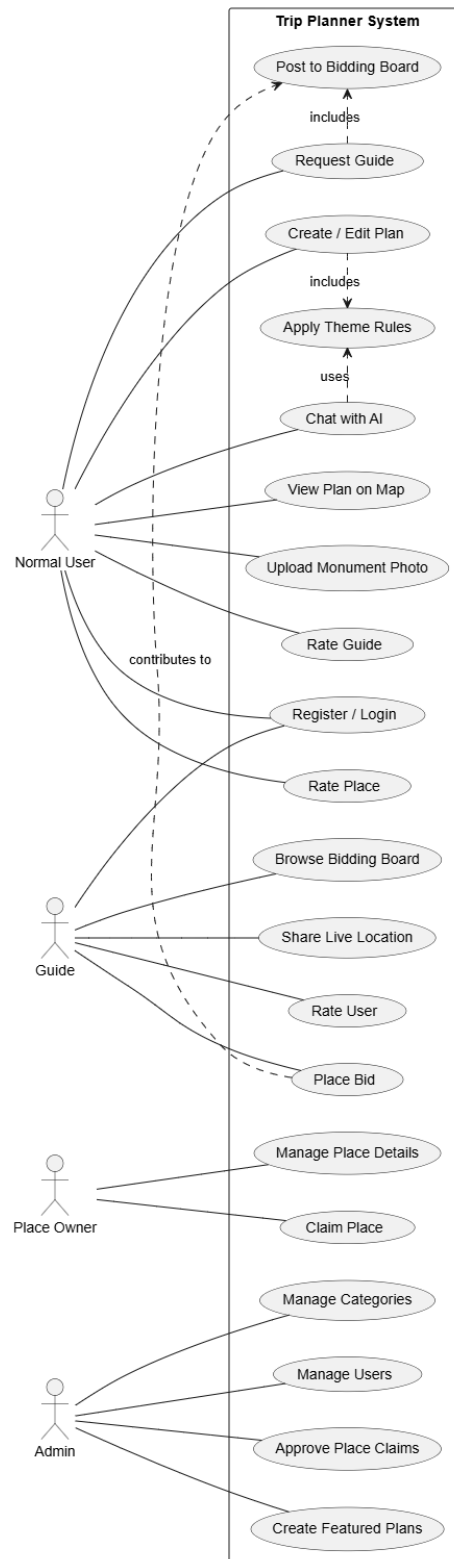


Figure 3-1 Use Case Diagram

3.5 Use Case Tables

Table 3-1 UC-1 Register Account

Use Case Identification	
Use Case ID:	UC-1
Use Case Name:	Register Account
User/Actor:	User

Preconditions
- User is not registered in the system.

Basic Flow		
Step	User Actions	System Actions
1	User selects "Sign Up"	
2	User enters required information (email, phone, password)	System validates the data.
3		System creates the account and sends the user back to the login page
4	User Enters required credentials for login	System logs the user in.
Alternate Flow		
Step	User Actions	System Actions
3a	User Entered an Email or a phone that already exists	System Displays error message

Post conditions
- A new user account is created. - User profile is initialized.

Table 3-2 UC-2 Login

Use Case Identification	
Use Case ID:	UC-2
Use Case Name:	Login
User/Actor:	User

Preconditions
- User has a registered account.

Basic Flow		
Step	User Actions	System Actions
1	User enters email, phone and password.	System validates the data.
2		System checks if the user exists and logs the user after.
Alternate Flow		
Step	User Actions	System Actions
2a	User Entered invalid credentials	System Displays error message and access is denied

Post conditions
- User is authenticated and logged in.

Table 3-3 UC-3 Generating an AI trip plan

Use Case Identification	
Use Case ID:	UC-3
Use Case Name:	Generating an AI Trip Plan
User/Actor:	User

Preconditions
- User is logged in.

Basic Flow		
Step	User Actions	System Actions
1	User selects theme (romantic, family, solo, etc.).	
2	User selects preferences (budget, locations).	System sends preferences and themes selected to the AI.
3		System generates a plan with places.
4		Plan is displayed on map.
Alternate Flow		
Step	User Actions	System Actions
3a	No matching results.	System suggests other plans with similar preferences.

Post conditions
- A generated plan is created and saved.

Table 3-4 UC-4 Join a Generated plan from other users

Use Case Identification	
Use Case ID:	UC-4
Use Case Name:	Join a Generated Plan From other Users
User/Actor:	User

Preconditions
- User is logged in. - Generated plan exists. - Plan has available capacity.

Basic Flow		
Step	User Actions	System Actions
1	User views public generated plans.	
2	User selects a plan.	System gives all details on selected plan, like theme, preferences and places
3	User clicks "Join".	System confirms participation.
Alternate Flow		
Step	User Actions	System Actions
3a	User clicks "Create Group" to start his own group with no existing participants	System Creates a new group with the same plan and adds it to the list of groups under the plan

Post conditions
- User is added as participant.

Table 3-5 UC-5 Guide Bidding and Assignment

Use Case Identification	
Use Case ID:	UC-5
Use Case Name:	Guide Bidding and Assignment
User/Actor:	Plan Owner, Guide

Preconditions
- A generated plan exists. - Participant threshold is met. - Plan owner has requested a guide.

Basic Flow		
Step	User Actions	System Actions
1	Guide views available guide-requested plans.	
2	Guide selects a plan.	System sends plan details to the guide.
3	Guide enters bid price and optional message.	System validates guide eligibility.
4		System stores the bid.
5	Plan owner views submitted bids.	
6	Plan owner selects one bid.	System assigns the selected guide to the plan.
7		System automatically rejects all other bids.
8		System locks the plan and notifies all involved participants.

Alternate Flow		
Step	User Actions	System Actions
3a	Guide not verified	Bid rejected.
6a	Owner Cancels process	No Guide Assigned

Post conditions
- One guide is assigned to the plan. - Other bids are automatically rejected. - Plan becomes locked for modification.

Table 3-6 UC-6 Landmark Recognition

Use Case Identification	
Use Case ID:	UC-6
Use Case Name:	Landmark Recognition
User/Actor:	User

Preconditions
- User has camera access enabled.

Basic Flow		
Step	User Actions	System Actions
1	User takes photo of landmark and sends it to the AI	
2		System analyzes image using AI.
3		System identifies landmark and displays historical and cultural information
Alternate Flow		
Step	User Actions	System Actions
3a	Landmark is not recognized	System suggests retry

Post conditions
- Landmark information is displayed.

Table 3-7 UC-7 Place Rating

Use Case Identification	
Use Case ID:	UC-7
Use Case Name:	Place Rating
User/Actor:	User

Preconditions
- User visited place through plan.

Basic Flow		
Step	User Actions	System Actions
1	User selects place.	
2	User submits rating and comment.	System stores review.
3		System updates the place rating.

Post conditions
- Review is stored and visible.

Table 3-8 UC-8 Create Featured Plan

Use Case Identification	
Use Case ID:	UC-8
Use Case Name:	Create Featured Plan
User/Actor:	Guide / Company

Preconditions
- Guide or company are verified.

Basic Flow		
Step	User Actions	System Actions
1	Guide enters plan details and places	
2	Guide submits plan	System validates the details and publishes the plan
Alternate Flow		
Step	User Actions	System Actions
1a	Missing required information from plan details	Error displayed and user is sent back to fill the form again.

Alternate Flow		
Step	User Actions	System Actions
1b	Company enters plan details and places	
2b	Company submits plan and assigns a guide to it	System validates the details and publishes the plan.

Post conditions
- Plan is created and published.

3.6 UML Class Diagram

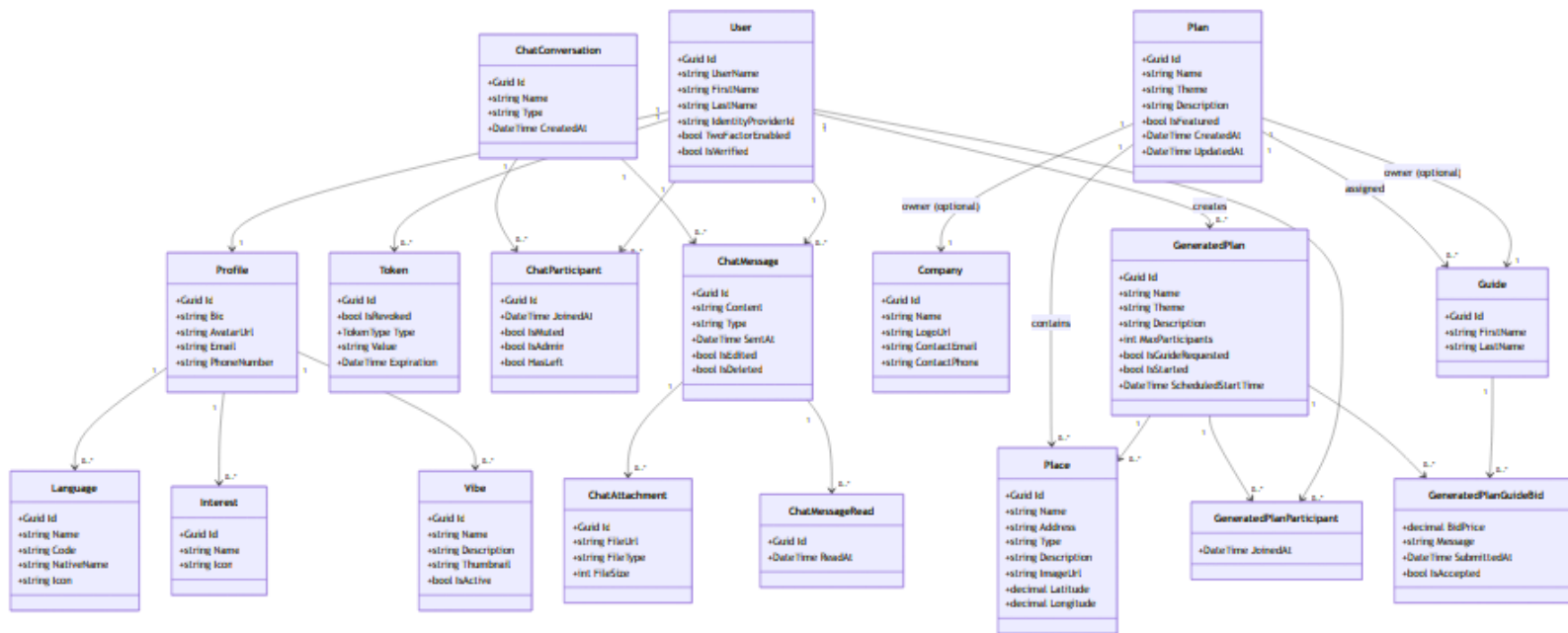


Figure 3-2 UML Class Diagram

3.7 ERD Diagram

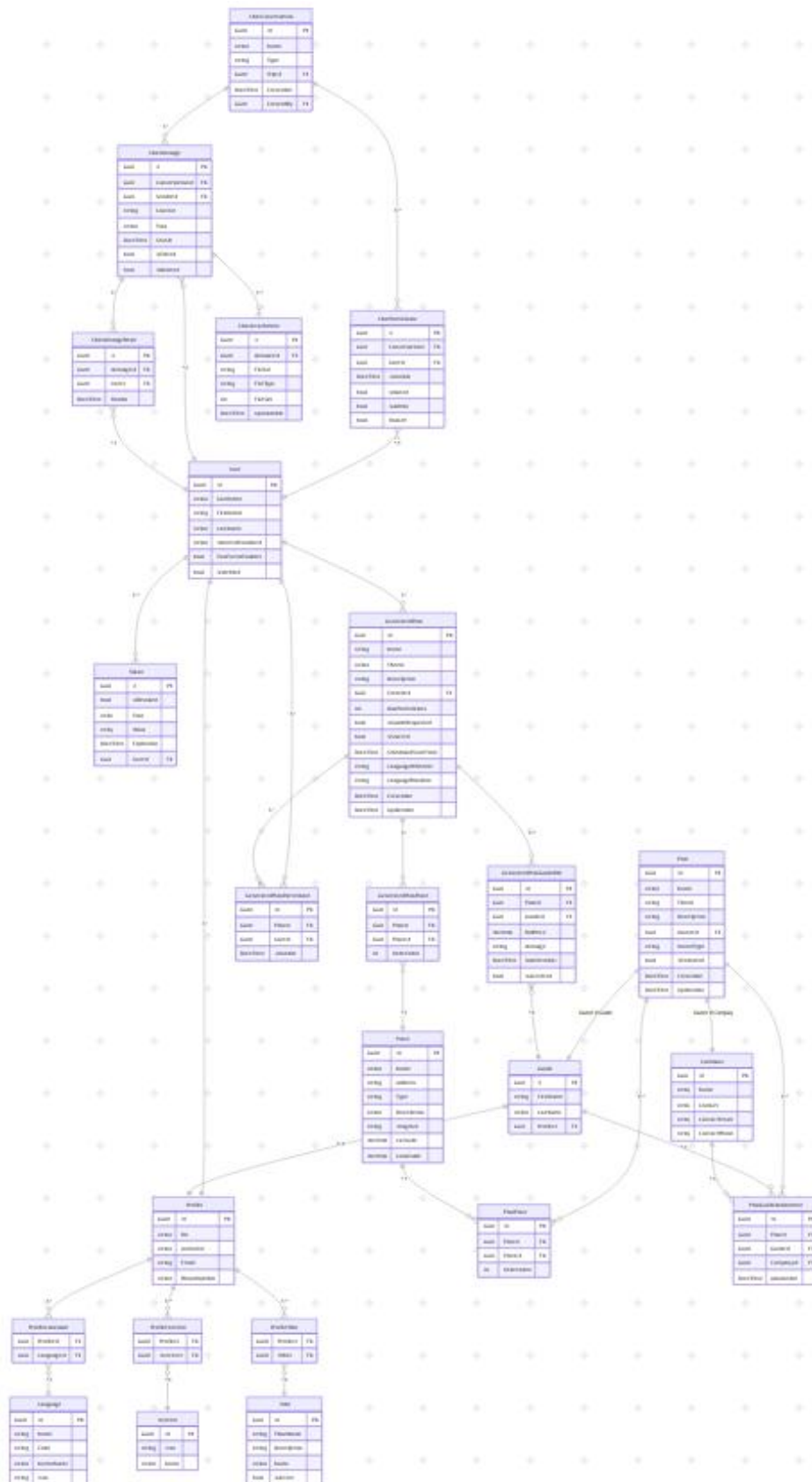


Figure 3-3 ERD Diagram

3.8 Sequence Diagrams

- User Generates AI Trip Plan

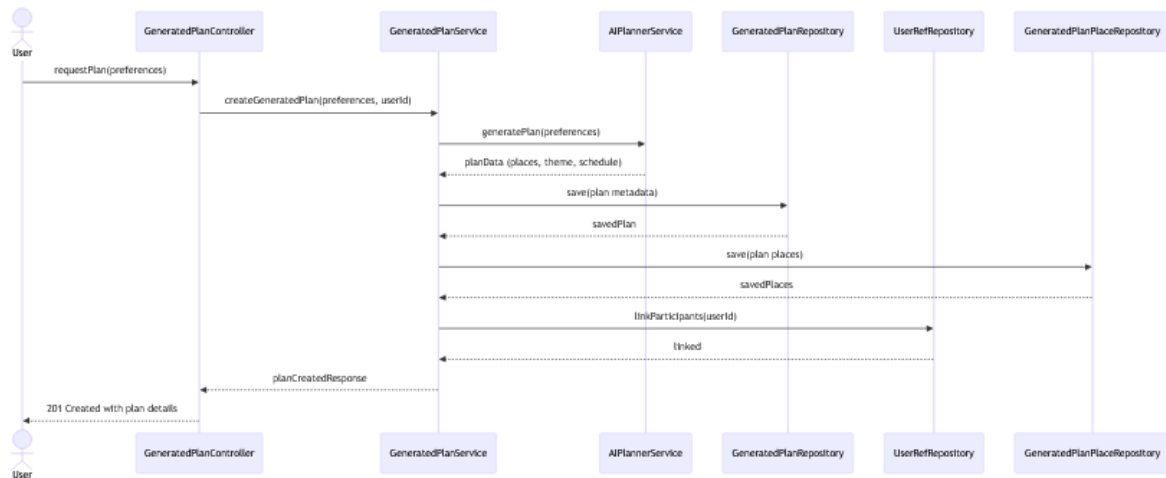


Figure 3-4 Sequence Diagram for User Generating an AI Trip Plan

- User Requests a Guide for Plan (Bidding)



Figure 3-5 Sequence Diagram for User Requesting a Guide for Plan

- Monument Recognition via AI

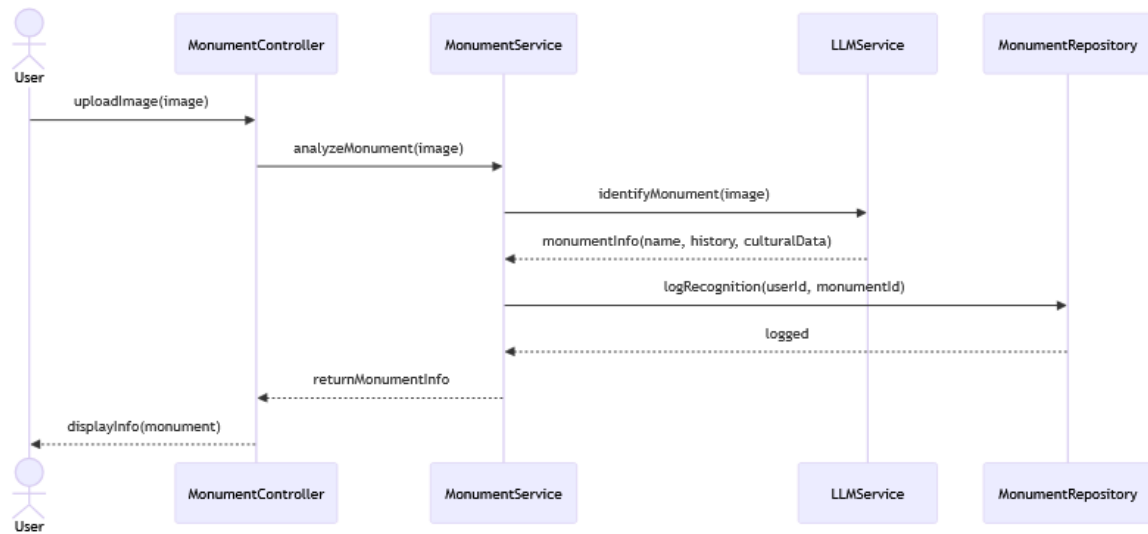


Figure 3-6 Sequence Diagram for Monument Recognition via AI

- User Joins a Generated Plan

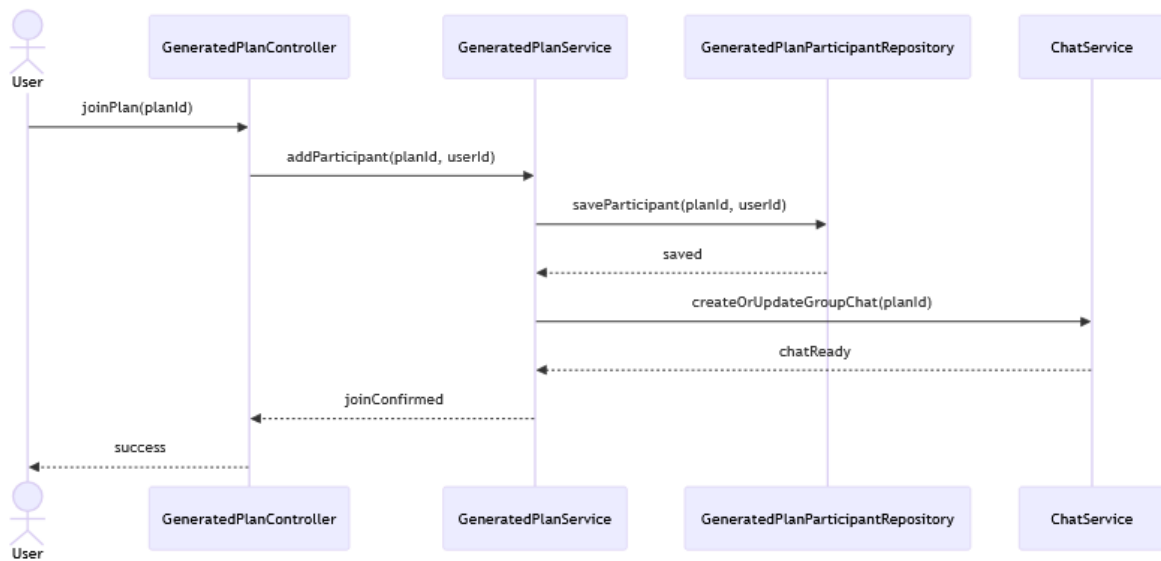


Figure 3-7 Sequence Diagram for User Joining a Generated Plan

4 AI Methodolgy

4.1 Data Collection

The TOURA platform relies on two distinct but complementary datasets: historical data for artifact recognition and cultural information, and planning data for trip itinerary generation and service recommendations. This section focuses specifically on the collection and preparation of historical data, which forms the foundation of the interactive museum experience component.

4.1.1 Historical Data Collection Strategy

The historical dataset serves a dual purpose: training computer vision models to recognize Egyptian monuments and artifacts, and providing rich historical narratives for conversational AI to deliver to tourists. Unlike generic image classification tasks, Egyptian artifact recognition requires both visual discrimination and deep cultural-historical knowledge integration.

4.1.2 Image Dataset Composition

The visual recognition dataset comprises **75 distinct monuments and statues**, each represented by a minimum of 15 images, resulting in over 1,100 total training images. This dataset structure was deliberately designed to balance coverage breadth with per-class sample sufficiency.

Dataset Characteristics:

Per-artifact image count: Minimum 15 images per monument/statue

- **Total unique artifacts:** 75 historical objects
- **Artifact diversity:** Includes statues of pharaohs (Ramses II, Tutankhamun, Cleopatra, etc.), architectural monuments (temples, pyramids, obelisks), funerary objects (sarcophagi, canopic jars), and decorative artifacts (jewelry, pottery, wall reliefs)
- **Image variations:** Multiple angles, lighting conditions, distances, and photographic contexts to ensure model robustness

The decision to require at least 15 images per artifact reflects computer vision best practices for few-shot learning scenarios. While deep learning models typically benefit from thousands of examples per class, transfer learning from pre-trained vision models enables

effective fine-tuning with more modest datasets when supplemented with appropriate augmentation strategies.

4.1.3 Data Source and Collection Process

Data collection followed a rigorous manual curation process prioritizing authenticity and quality over quantity:

Collection Methodology:

1. **Museum-focused sourcing:** Images were collected exclusively from artifacts housed in Egyptian museums, including the Egyptian Museum in Cairo, Luxor Museum, Aswan Museum, and other national heritage institutions. This geographic constraint ensures that recognized artifacts are actually accessible to tourists using the platform.
2. **Manual search and verification:** Each artifact was identified through manual research on Egyptian pharaohs, dynasties, and historical periods. When a significant artifact was identified, the team searched for its current museum location and collected photographic representations.
3. **Multi-source image gathering:** Images were sourced from:
 - Official museum digital collections
 - Archaeological databases and academic publications
 - High-quality tourism photography archives
 - Educational resources from Egyptological institutions
4. **Quality filtering:** Only clear, well-lit images showing sufficient detail for visual recognition were included. Blurry, heavily cropped, or contextually ambiguous images were excluded.

This manual approach, while labor-intensive, ensures dataset quality and relevance. Every artifact in the dataset represents a real object that tourists may encounter, and every image reflects realistic photography conditions.

4.1.4 Data Augmentation Strategy

Given the relatively modest dataset size (15+ images per artifact), data augmentation techniques were employed to artificially expand the training set and improve model generalization:

Augmentation Techniques Applied:

1. **Color adjustment:** Modifying brightness, contrast, and saturation to simulate different museum lighting conditions (natural light, artificial lighting, flash photography).
2. **Brightness variation:** Adjusting image brightness to account for underexposed or overexposed tourist photographs.
3. **Selective augmentation:** Augmentation was applied primarily to artifacts with fewer source images, ensuring all classes achieved comparable effective sample sizes.

4.2 Model Research & Selection

The TOURA platform integrates multiple AI models to deliver its comprehensive functionality. This section details the research, evaluation, and selection process for the core models powering artifact recognition and conversational interaction.

4.2.1 Multimodal Embedding Model for Visual-Semantic Understanding

Challenge: The artifact recognition task requires matching visual input (tourist photographs) with semantic historical information (text narratives). Traditional computer vision models perform image classification but lack semantic understanding, while language models lack visual processing capabilities.

Solution Approach: Multimodal embedding models that project both images and text into a shared embedding space, enabling semantic similarity matching between visual and textual content.

Selected Model: CLIP (Contrastive Language-Image Pre-training)

After extensive research and experimentation, the project selected **CLIP-ViT-L-14** from the Sentence Transformers library:

python

```
CLIP_MODEL_NAME = "sentence-transformers/clip-ViT-L-14"
```

```
clip = SentenceTransformer(CLIP_MODEL_NAME, cache_folder=MODEL_DIR)
```

Model Characteristics:

- **Architecture:** Vision Transformer (ViT) with Large configuration (14x14 patch size)
- **Training paradigm:** Contrastive learning on 400 million image-text pairs
- **Embedding dimension:** 768-dimensional dense vectors
- **Capability:** Joint embedding of images and text in a unified semantic space

Selection Rationale:

CLIP was chosen over alternative approaches for several compelling reasons:

1. **Multimodal capability:** Unlike traditional CNNs trained only on image classification, CLIP understands both visual content and textual descriptions, enabling semantic search where tourist photos can be matched against historical narratives.
2. **Zero-shot transfer:** CLIP's pre-training on diverse internet data provides strong baseline performance on Egyptian artifacts without requiring massive domain-specific datasets.
3. **Semantic understanding:** The model captures conceptual relationships (e.g., understanding that a "pharaoh statue" relates to "royal authority," "ancient Egypt," "divine kingship") beyond simple visual pattern matching.
4. **Robustness:** Contrastive training makes CLIP robust to variations in image quality, lighting, and perspective—critical for handling diverse tourist photography.
5. **Established performance:** CLIP has demonstrated state-of-the-art results on visual-semantic tasks across multiple benchmarks.

Alternative Models Evaluated

Several alternative embedding approaches were tested during development:

1. Traditional CNN-based embeddings (ResNet, EfficientNet):

- **Weakness:** Pure vision models lack semantic understanding; they classify images into predefined categories but cannot match against text descriptions
- **Result:** Poor performance on nuanced artifact queries and inability to handle artifacts not in predefined classes

2. Smaller CLIP variants (CLIP-ViT-B/32):

- **Weakness:** Lower embedding dimension and reduced model capacity led to weaker semantic discrimination
- **Result:** Acceptable performance but noticeably inferior to ViT-L-14 on artifacts with subtle visual differences

3. Custom-trained Siamese networks:

- **Weakness:** Requires large paired image-text datasets; limited generalization
- **Result:** Development time and data requirements exceeded project scope

Experimental Comparison:

Informal testing on a validation set of 150 artifact images showed:

- CLIP-ViT-L-14: ~87% top-1 retrieval accuracy, ~96% top-3 accuracy
- CLIP-ViT-B/32: ~78% top-1 retrieval accuracy
- ResNet-50 embeddings: ~62% top-1 retrieval accuracy (classification-only)

These results confirmed CLIP-ViT-L-14 as the optimal choice for the project's requirements.

4.2.2 Large Language Model Selection

The conversational AI component requires a large language model capable of understanding tourist queries, accessing historical knowledge, and generating engaging, accurate responses.

Selected Models: Gemini and GPT-4

TOURA employs a dual-LLM strategy, supporting both **Google Gemini** and **OpenAI GPT-4**:

Why Gemini and GPT-4:

1. **State-of-the-art language understanding:** Both models represent the current frontier in natural language processing, with superior reasoning, context understanding, and instruction-following compared to smaller or earlier models.
2. **Multimodal capability:** Both support native image understanding, enabling the LLM to directly process tourist photographs and reason about visual content alongside text queries.
3. **Long context windows:**
 - GPT-4: Up to 128K tokens (GPT-4 Turbo)
 - Gemini: Up to 1M tokens (Gemini 2.5 Pro)
 - This enables maintaining extensive conversation history and incorporating large knowledge bases directly in context.
4. **API accessibility:** Both provide robust, production-ready APIs with:
 - High availability and reliability
 - Comprehensive documentation and developer support
5. **Tool/function calling:** Both models support structured function calling, essential for LangGraph tool integration (Google Lens, web search, database queries).
6. **Safety and content filtering:** Built-in safeguards prevent generation of inappropriate or harmful content, important for public-facing tourism applications.

4.2.3 LangGraph Architecture and Tool Integration

The conversational system is built using **LangGraph**, a framework for creating stateful, multi-agent LLM applications:

Core Architecture Components:

1. **State management:** LangGraph maintains conversation history, user preferences, and context across multi-turn interactions

2. **Memory systems:**

- **Short-term memory:** Recent conversation turns for immediate context
- **Long-term memory:** User preferences, trip details, and interaction history

3. **Tool integration:** The LLM agent has access to multiple tools:

a. Google Lens API:

- **Purpose:** Fallback visual recognition when CLIP cannot confidently identify an artifact
- **Usage:** If an artifact is not in the TOURA database, the system queries Google Lens for identification
- **Integration:** Results are parsed and presented to the LLM for response generation

b. SerpAPI (Web Search):

- **Purpose:** Retrieve current information not in the static knowledge base
- **Usage:** When users ask complex, current, or niche questions beyond the historical database

c. FAISS Vector Database:

- **Purpose:** Semantic search over historical narratives
- **Usage:** Retrieve relevant historical context based on query embeddings

- **Implementation:** Pre-indexed artifact descriptions enable fast retrieval

4. Conversation flow management:

- **Query routing:** Determines whether a question requires visual recognition, database lookup, or web search
- **Response synthesis:** Combines information from multiple sources into coherent answers

Clarification handling: Asks follow-up questions when user intent is ambiguous

5 TouRA Design

5.1 UI UX

- Login and register screens

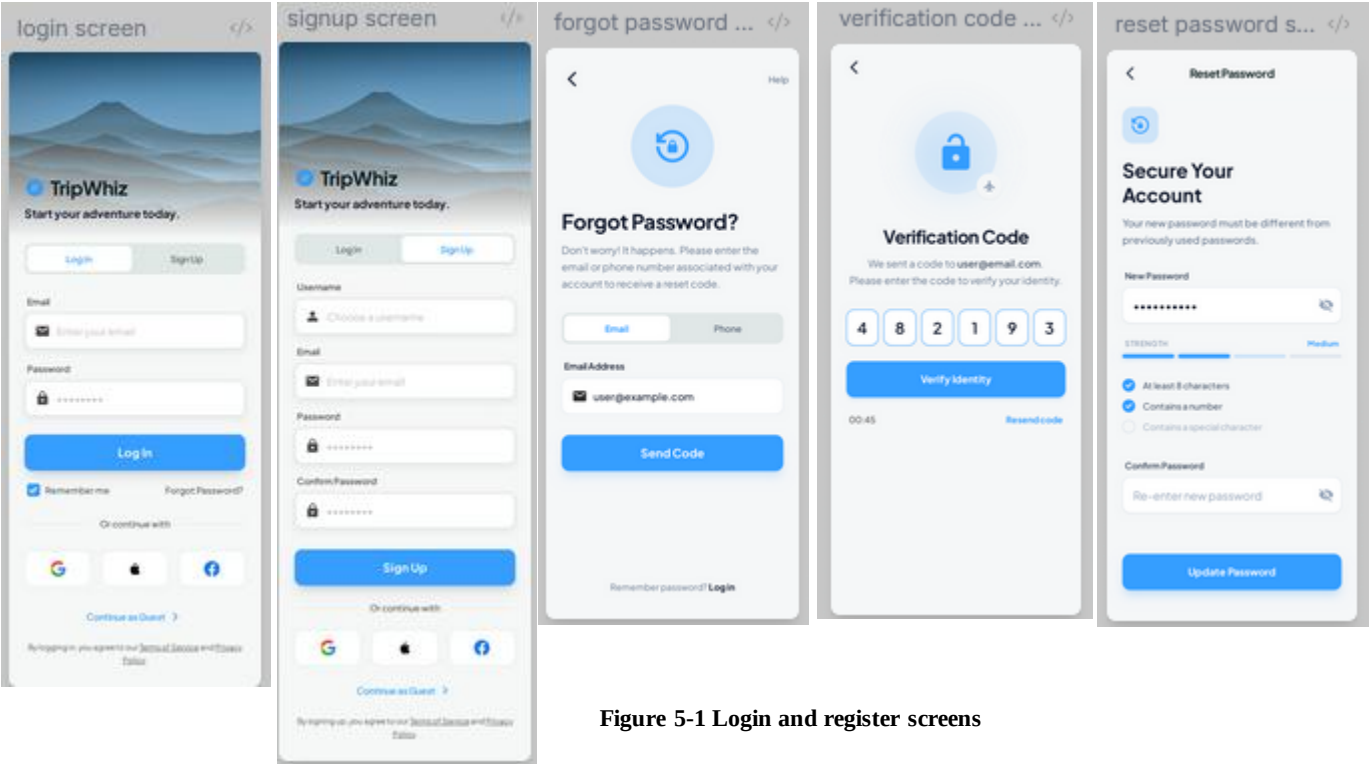


Figure 5-1 Login and register screens

- AI Chat

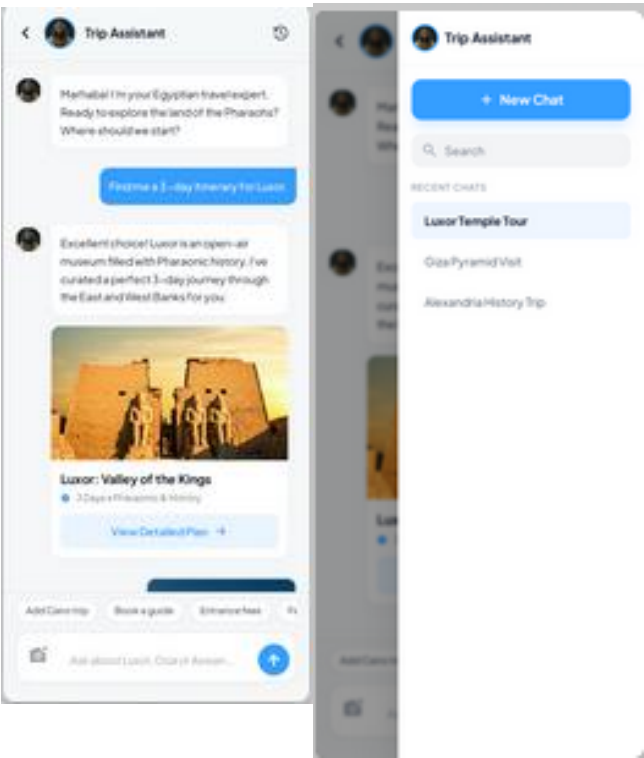


Figure 5-2 AI Chat

- Chat with a guide

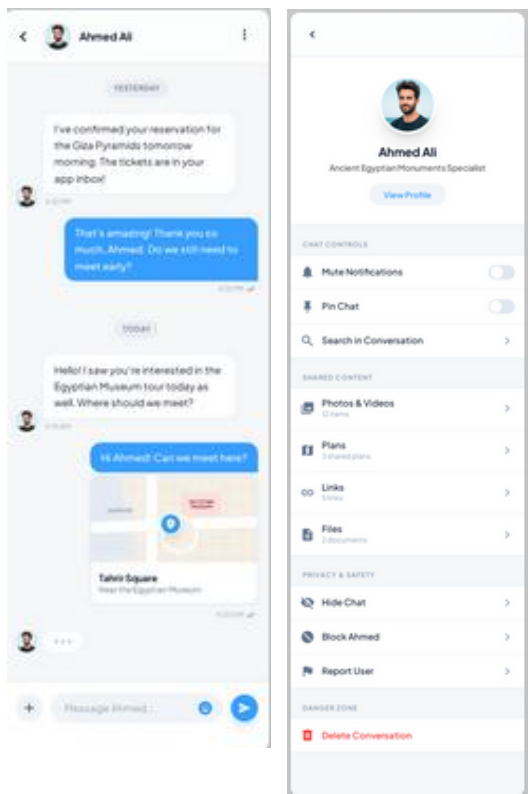


Figure 5-3 Chat with a guide

- Trip Group Chat

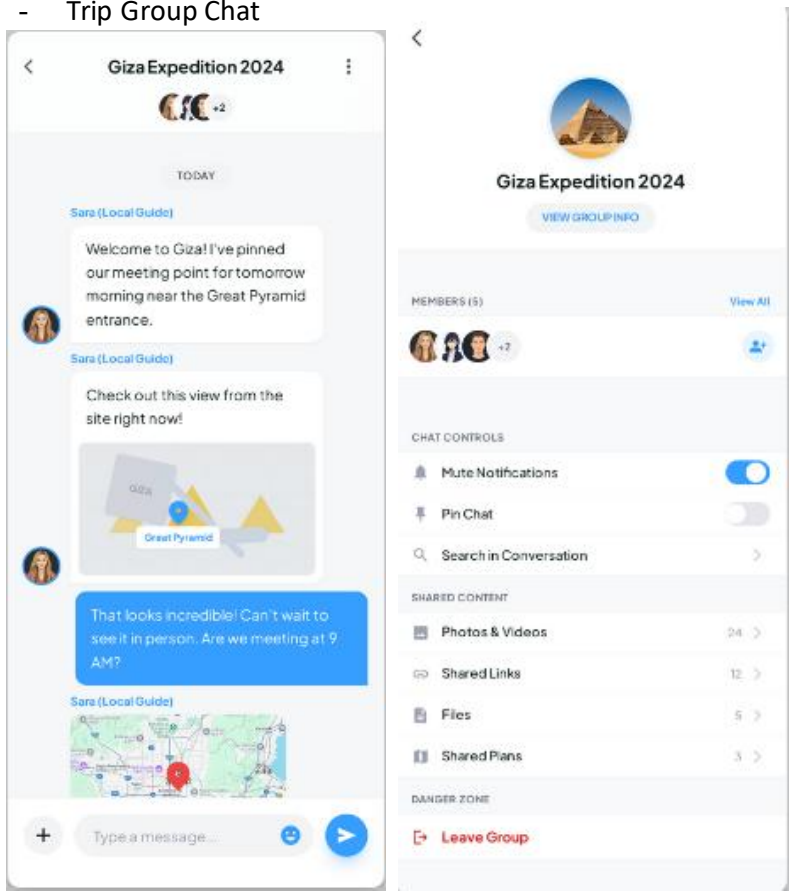


Figure 5-4 Trip group chat

- Profile Setup

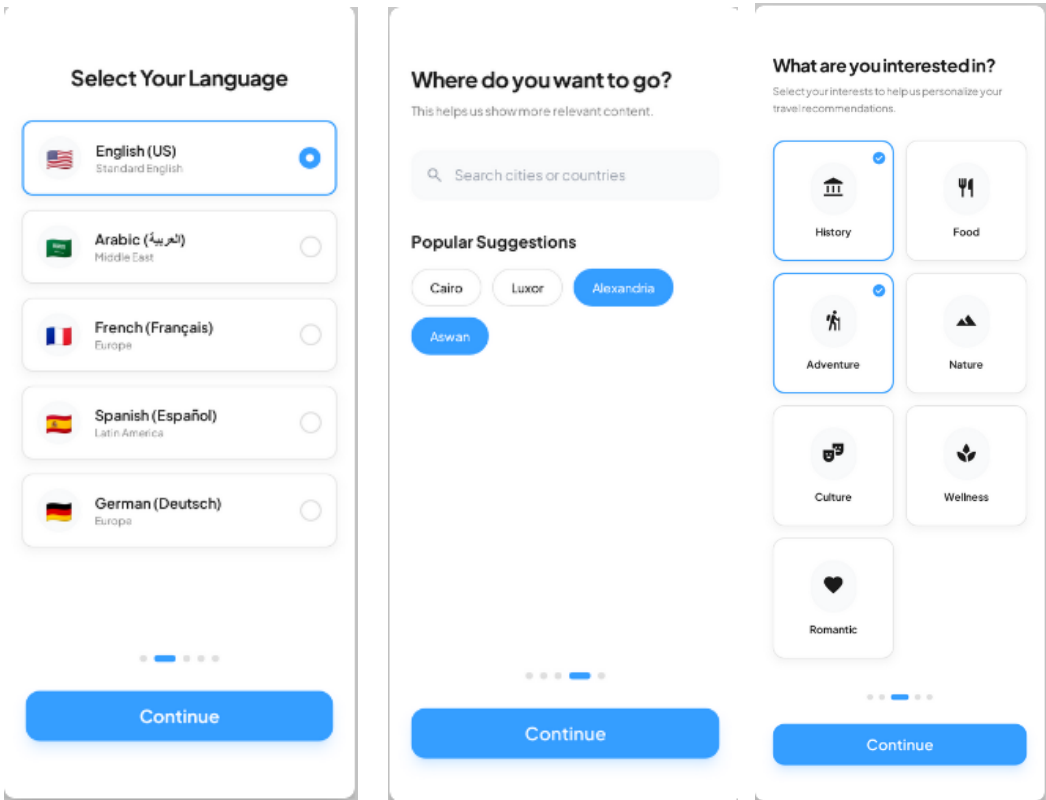


Figure 5-5 Profile setup screens

- Payment and billings

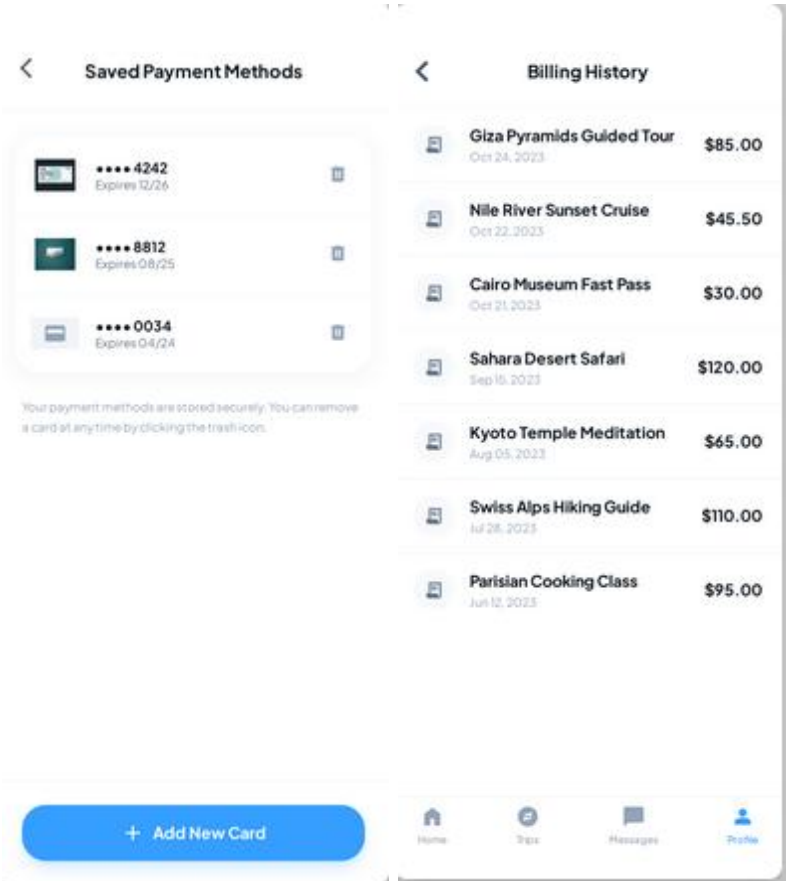


Figure 5-6 Payment and billings screens

- Trip Exploration

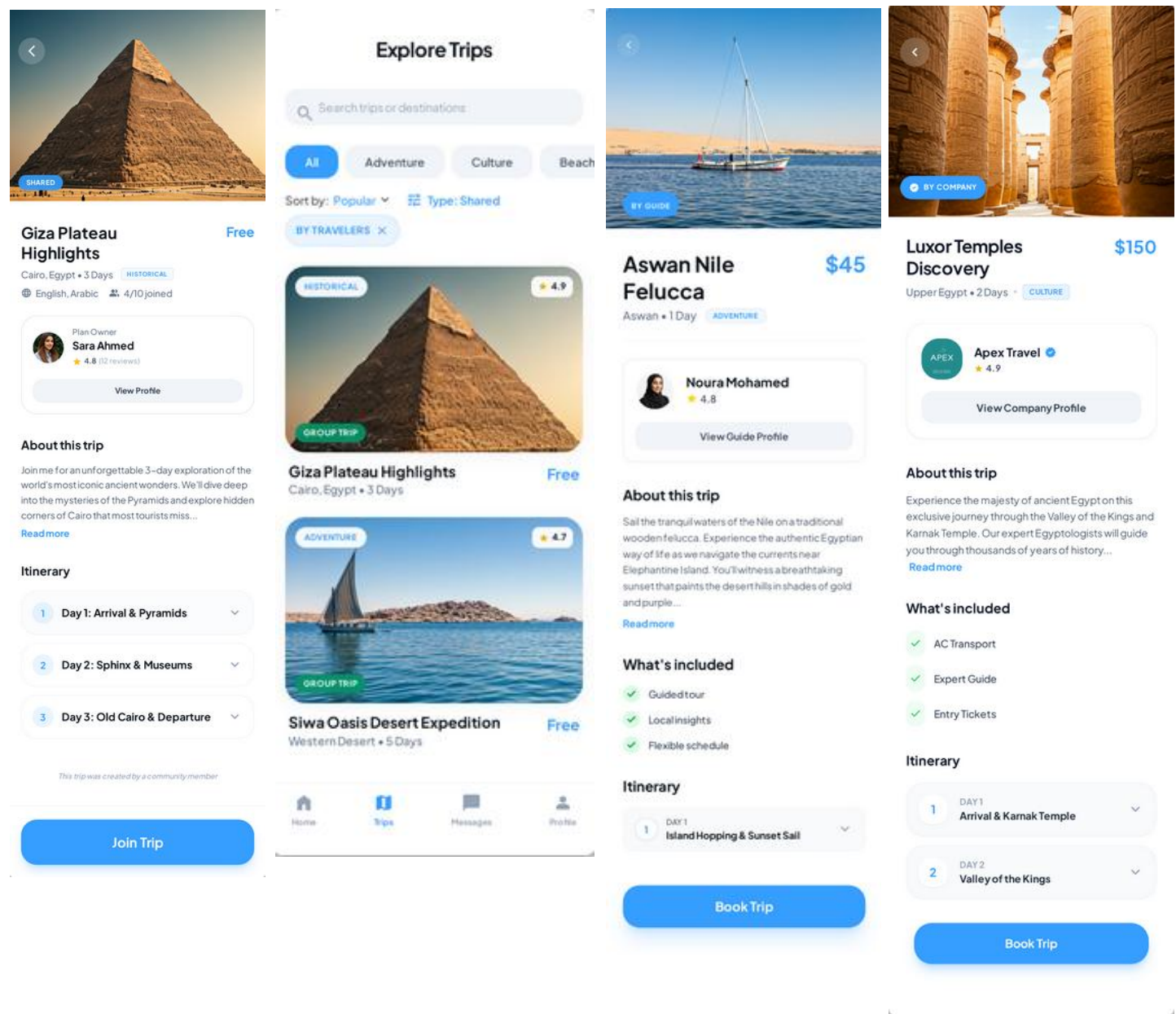


Figure 5-7 Trip Exploration screens

- Plan Creation

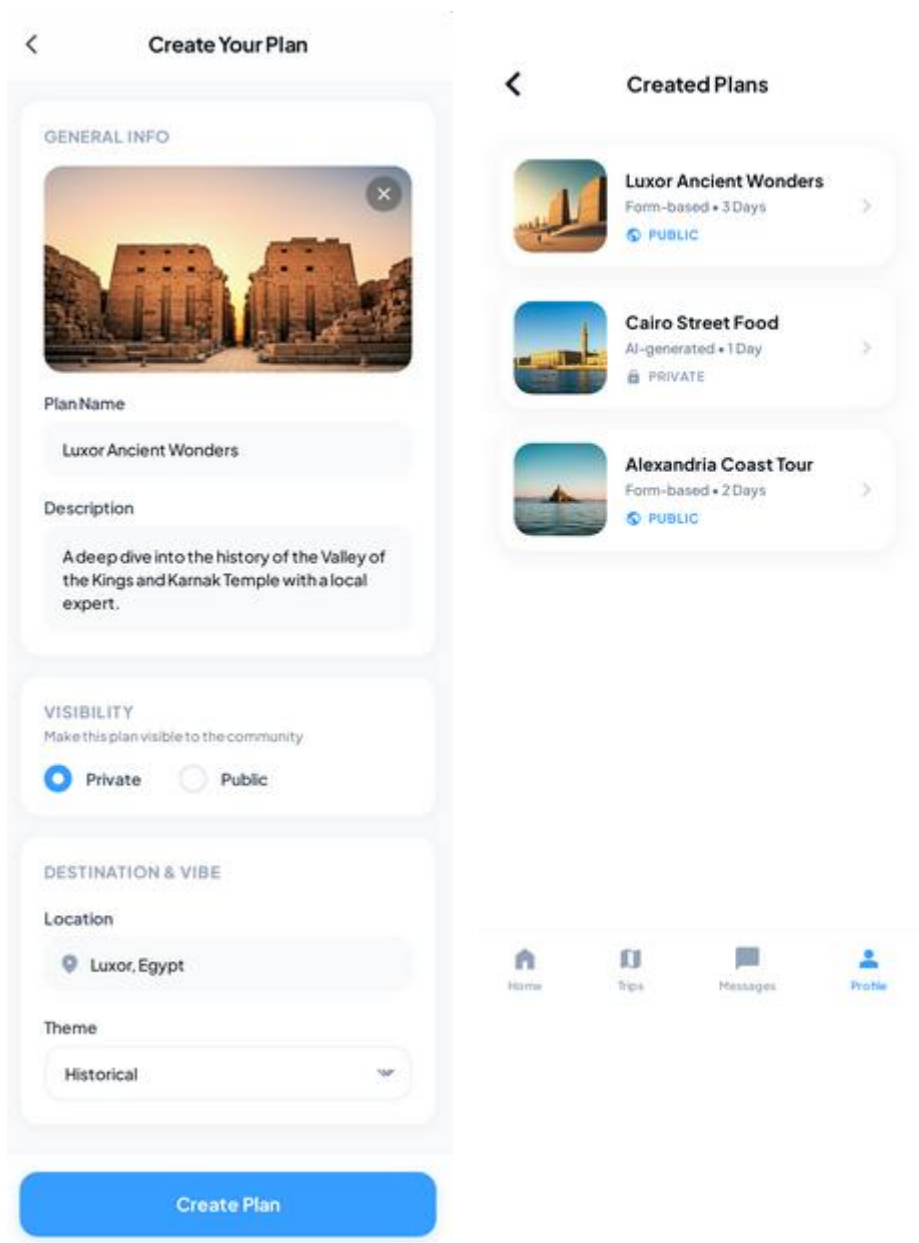


Figure 5-8 Plan Creation Screens

- Profile Details

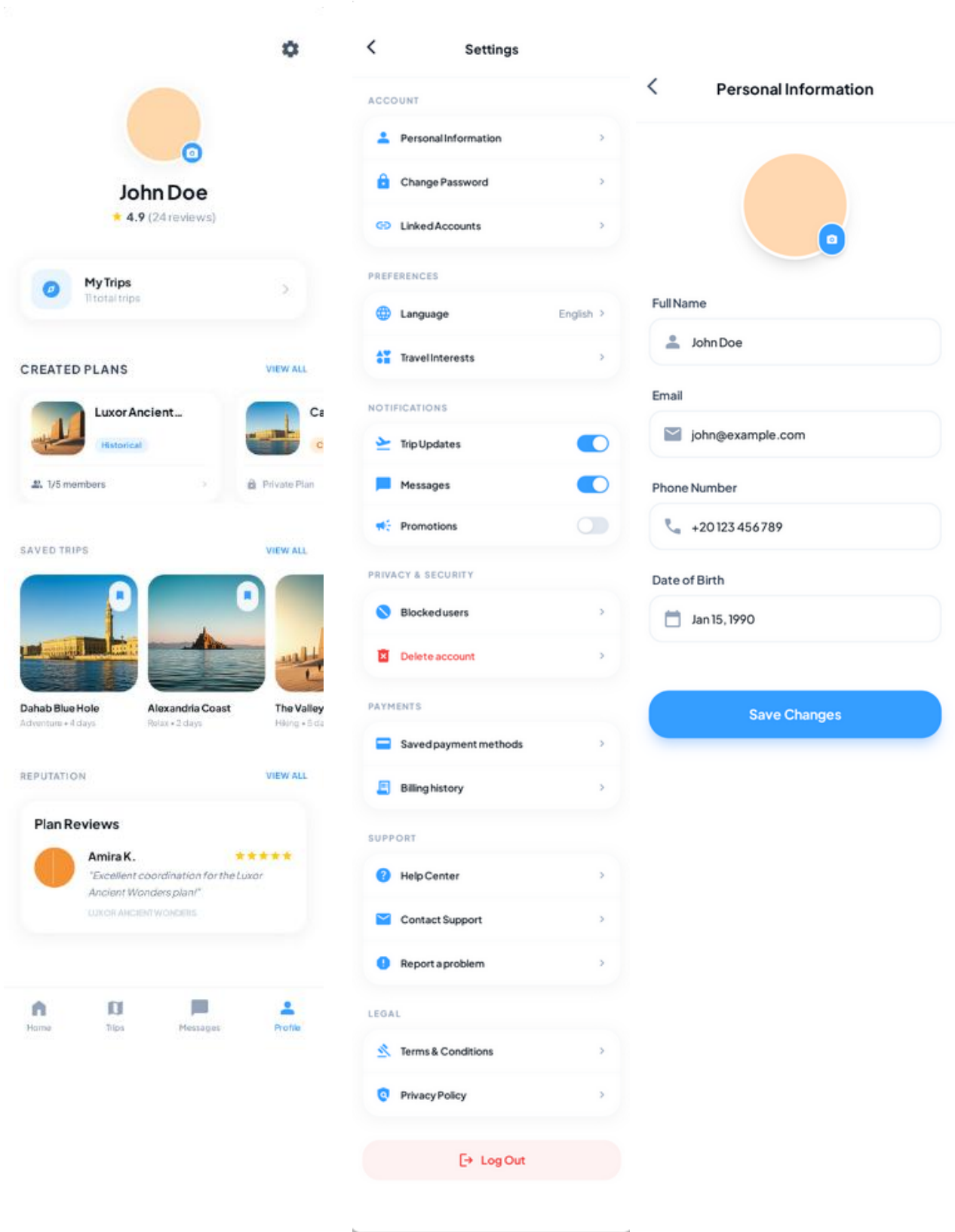


Figure 5-9 Profile Details Screens

- More on profile details

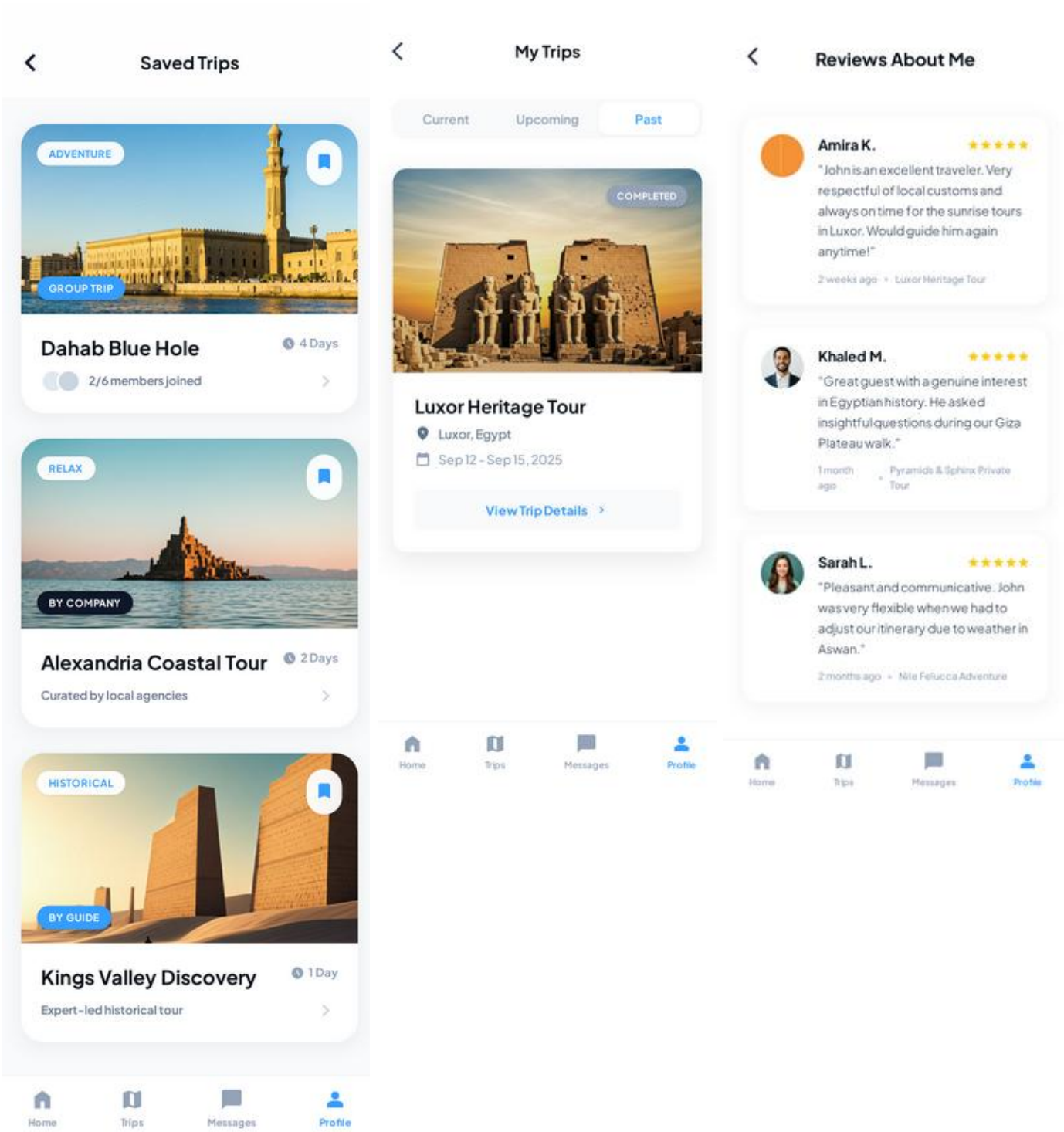


Figure 5-10 More profile screens

- Companies and company created plans

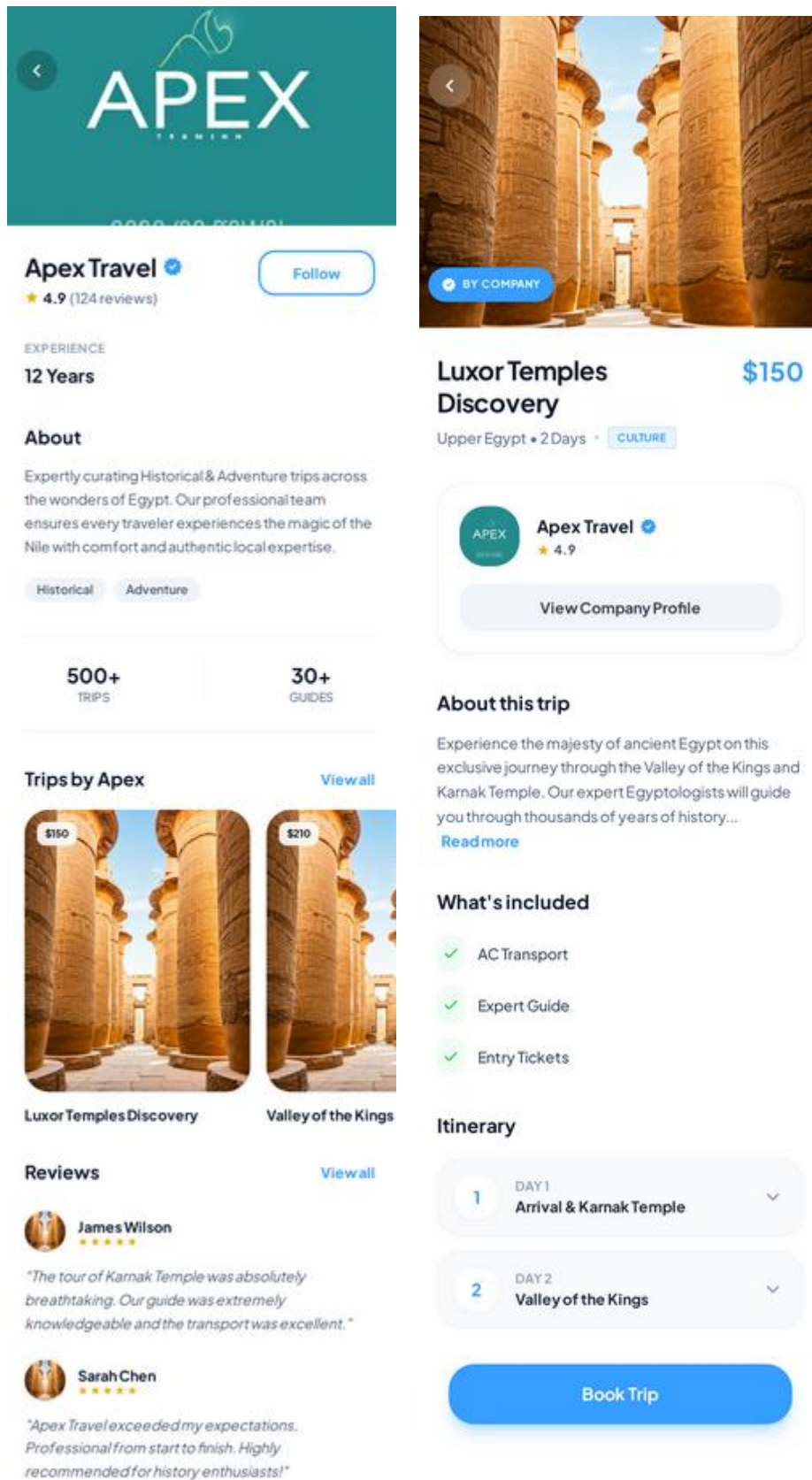


Figure 5-11 Company screens

- Home Screen

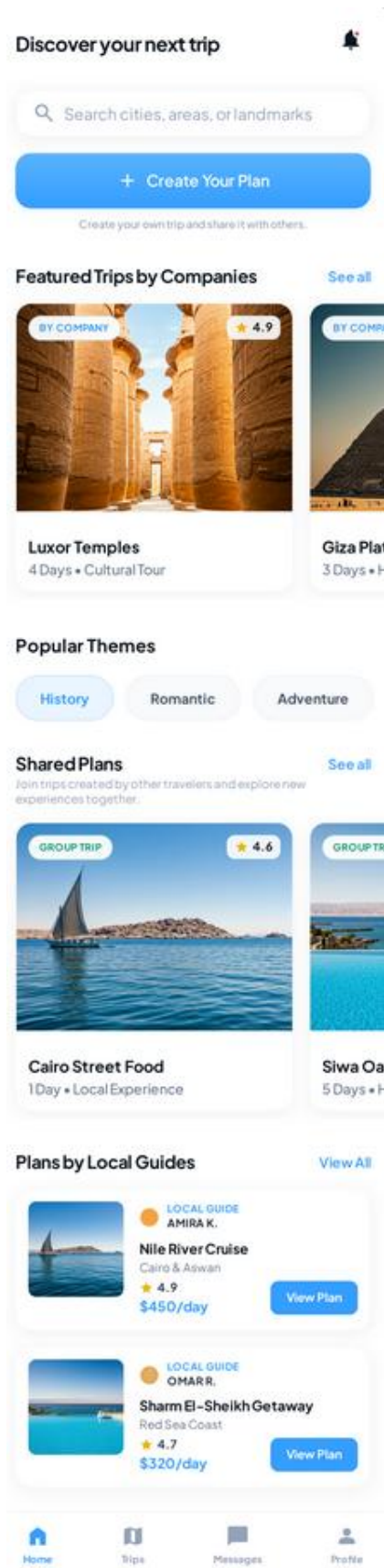


Figure 5-12 Home Screen

- Messages screen

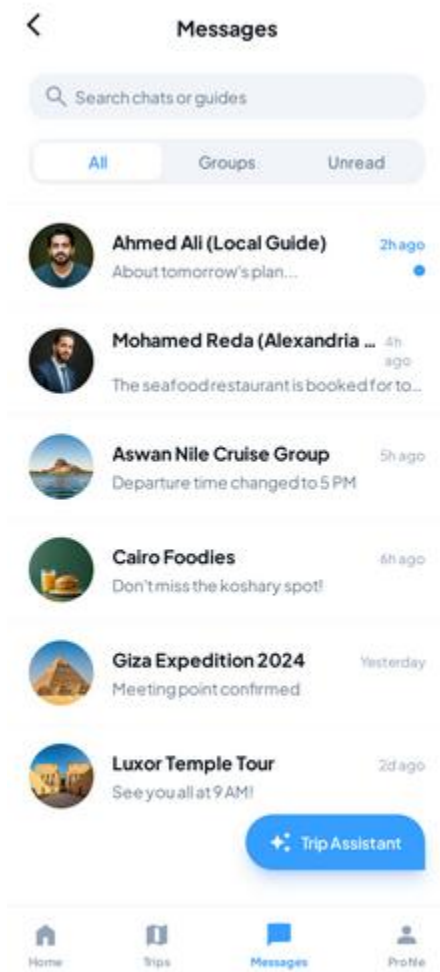


Figure 5-13 Messages Screen

- Notifications screen

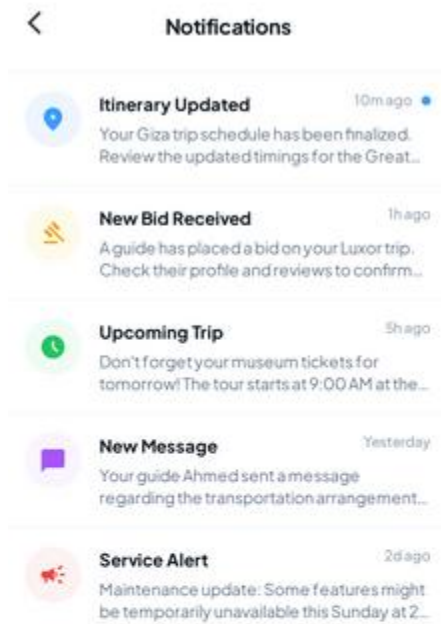


Figure 5-14 Notifications screen

- Map view for a plan



Figure 5-15 Map view Screens

- Search Screen

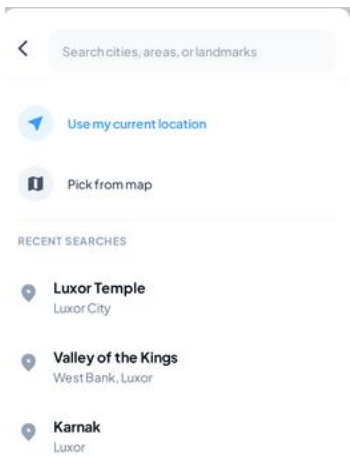


Figure 5-16 Search Screen

- Confirm booking and payment screens

<

Confirm Booking



Luxor Temples Discovery

2 Days

by Apex Travel

NUMBER OF TRAVELERS

Travelers

−

2

+

PRICEDetails

Price per person

\$150

Total for 2 travelers

\$300

Confirm & Pay

<

Payment



Luxor Temples Discovery

Luxor, Egypt

Oct 24 - 25, 2026

2 Travelers

2 Days

PRICEBREAKDOWN

Trip Price

\$300.00

Service Fee

\$15.00

Taxes

\$5.00

Total Amount

\$320.00

PAYMENT METHOD



Visa ending in 4242

Expires 12/26

Change

❗

Cancellation Policy: Free cancellation up to 48 hours before the trip. Non-refundable after this period.

Pay Now

Figure 5-17 Booking and Payment Screens

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6 TouRA Implementation

This chapter describes in depth how TouRA was designed and implemented. It covers the overall system architecture, the internal structure of the backend modules, the cross-cutting mechanisms used to keep the system reliable and consistent, and the design of the web administration dashboard and the mobile application. It also summarizes the tools and programming languages used throughout the project, along with the reasoning behind each selection.

6.1 System Architecture Overview

TouRA is built as a set of independently structured modules — Users, Trips, Conversations, Notifications, and Payments — each responsible for a distinct area of the system’s functionality. Rather than a single monolithic codebase, every module is organized internally using Clean Architecture, and modules interact with one another only through explicit, versioned Contracts projects (for example, `IUsersContract`, `IConversationsContract`, `INotifyContract`, and `IPayContract`) instead of direct references between their internal layers. A shared Common library provides cross-cutting building blocks — domain events, exceptions, correlation-id tracking, the transactional outbox/inbox infrastructure, and shared presentation concerns — that every module reuses, which keeps behavior consistent across the system while avoiding duplicated infrastructure code.

The full system is containerized with Docker, and Docker Compose is used to orchestrate the application services together with their supporting infrastructure — databases, the message broker, and MinIO object storage — so that the entire platform can be brought up consistently for development, testing, and deployment.

6.2 Backend Module Design (Clean Architecture)

Each backend module follows the same four-layer structure, which keeps business logic isolated from technical details and makes every module easier to reason about and test in isolation:

- Domain layer: entities, value objects, and domain events (for example, Trip, TripBidding, Place, Conversation, User, Document), built around a shared Entity abstraction with no dependency on any external framework.
- Application layer: use-case orchestration through services (for example, TripService, BiddingService, ReviewService, UserService, ChatService, NotificationService), request/response DTOs, validators for incoming requests, mappers between entities and DTOs, and repository/unit-of-work abstractions that the domain and application layers depend on without knowing their concrete implementation.
- Infrastructure layer: Entity Framework Core DbContexts and entity configurations, concrete repository implementations, outbox/inbox background jobs, and external service clients (for example, RoutingEngineClient, TripPlanClient, SemanticModelClient) that isolate third-party and AI service integration details from the rest of the module.
- Presentation layer: versioned ASP.NET Core Web API controllers (for example, TripController, BiddingController, PlaceController, UserController, NotificationController, PaymentsController), SignalR hubs, and the integration-event handlers that connect the module to events published by other modules.

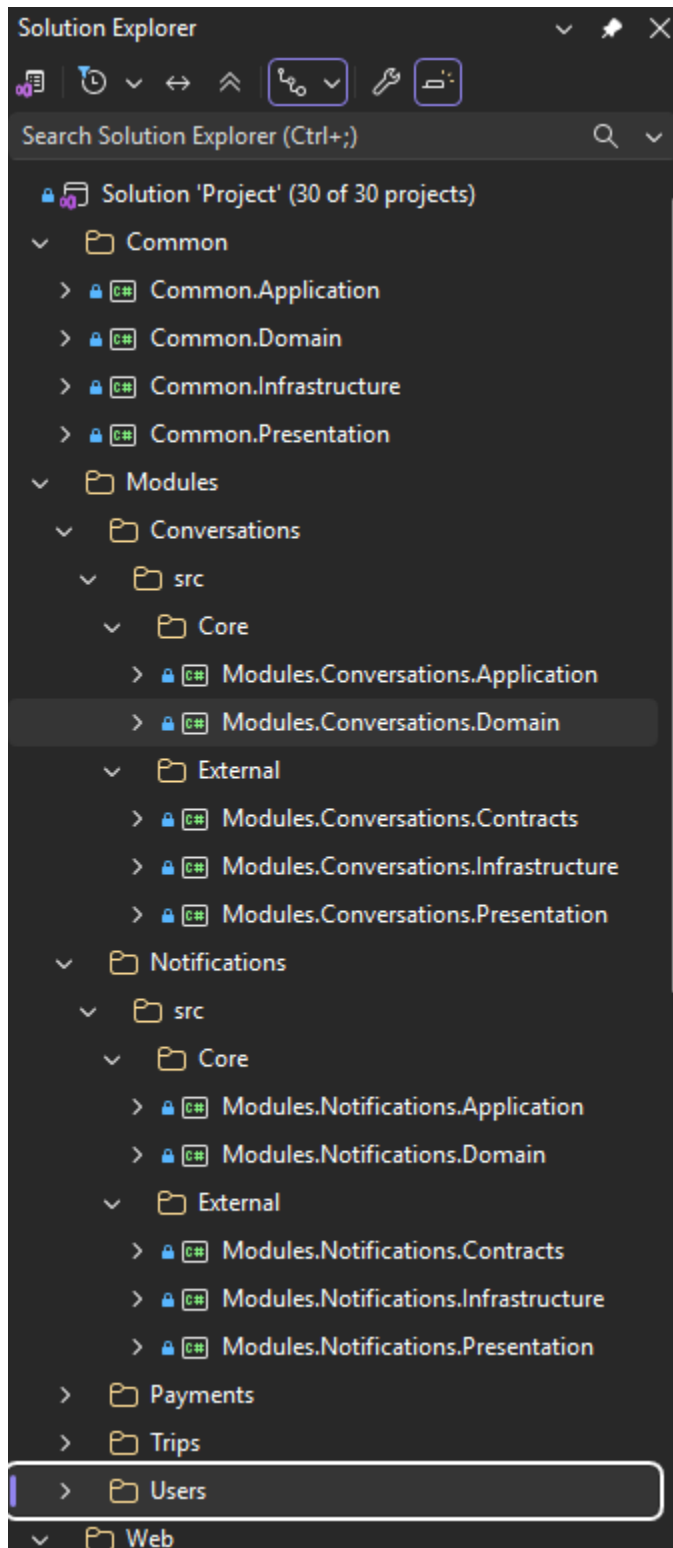


Figure 6-1 Backend Architecture

6.3 Reliable Messaging: Domain Events, Outbox, and Inbox

TouRA distinguishes between two kinds of events. In-process domain events (for example, `TripReadyDomainEvent`, `BiddingPlacedDomainEvent`, `TripParticipantReviewedDomainEvent`) are dispatched synchronously within the same transaction through a `DomainEventDispatcher`, allowing a module to react immediately to something that happened inside its own boundary, such as updating a projection or triggering a notification.

Communication between modules, by contrast, uses integration events delivered through a Transactional Outbox: whenever a module needs to notify another module of something (for example, a user being created, or a trip being ready), the event is written to an `OutboxMessage` table in the same database transaction as the business change, and a background `ProcessOutboxJob` later publishes it reliably to the shared event bus. On the receiving side, an Inbox mechanism (`InboxMessage`, `ProcessInboxJob`, and an idempotent-handler decorator) ensures that a redelivered message is not processed twice, even if the consumer or the network experiences a temporary failure. Together, these two patterns guarantee at-least-once, duplicate-safe communication between modules without requiring a distributed transaction.

6.4 Identity and Access Management

Authentication and authorization are handled internally by the Users module rather than through a third-party identity provider. Each account carries one or more roles (for example, User, Guide, Company, Place Owner, and Administrator) through a dedicated `UserRole` association, and a `TokenFactory` together with a `TokenService` issues and validates JWT access tokens once a user registers or logs in successfully. Role information is embedded directly in the token claims, which allows every module to authorize a request locally without a separate round trip to an identity service. A shared `ForwardedClaimsAuthenticationHandler` in the common presentation library then propagates these authenticated claims consistently to whichever module ultimately handles a given request. This self-contained, multi-role JWT approach was chosen over an external identity provider to keep the authentication flow simple, fast, and fully under the team's control, which was well suited to the project's relatively small and well-defined set of roles.

6.5 File Storage

User-generated content such as profile photos, place images, monument photographs submitted for AI recognition, and identity documents is stored in MinIO, an S3-compatible object storage service, through a shared `IFileService` abstraction. Keeping binary content in object storage rather than the relational databases keeps each module's database lean and makes it straightforward to serve and cache media independently of the application logic.

6.6 Real-Time Communication

Several parts of TouRA depend on immediate updates rather than periodic polling: trip and AI chat messages, system notifications, and the live location of an assigned guide. These are delivered through SignalR hubs (`ChatHub`, `NotificationHub`, and `TripHub`), which push updates directly to connected clients over a persistent WebSocket connection, supporting the real-time behavior described in the functional and non-functional requirements in Chapter 3.

6.7 AI and External Service Integration

The routing engine, the AI-assisted trip plan generator, and the artifact recognition and conversational features introduced in Chapter 4 are not implemented inside the core .NET modules. Instead, they run as independent Python microservices and are accessed from the `Trips` and `Conversations` modules through small, dedicated HTTP clients (`RoutingEngineClient`, `TripPlanClient`, and `SemanticModelClient`). Application code depends only on narrow interfaces such as `IRoutingService` and `IAiTripSuggestionService`, which keeps the core business logic independent of the specific AI stack behind these services and made it possible to substitute a `MockRoutingService` during local development and testing, without needing a live connection to the external AI services. The internal design of these two AI microservices — the Statue Storyteller and the AI Travel Planner — is described in the following section.

6.8 AI Microservices Implementation

Two Python-based microservices power TouRA's AI-driven features: the Statue Storyteller, which handles monument recognition and conversational historical narration, and the AI Travel Planner, which generates multi-agent, multi-day itineraries. Both are deployed and scaled independently of the .NET backend.

6.8.1 Implementation Environment and Setup

Table 6-1 Statue Storyteller technology stack.

Technology	Version / Configuration	Role
Python	3.11+	Primary backend language
FastAPI + Uvicorn	Latest stable	HTTP framework and ASGI server
SQLAlchemy	2.x (ORM)	Database models, session management, and queries
psycpg (psycpg3)	Latest stable	Async-compatible PostgreSQL driver
LangGraph	Latest stable	Agent graph construction and turn-by-turn execution
Groq SDK	Latest stable	LLaMA-based LLM inference (fast GPU inference via Groq LPU)
OpenRouter	REST API	Secondary LLM routing across multiple model backends
DINOv2	Meta Research checkpoint	Vision encoder for statue image feature extraction
Google Lens API	REST (unofficial interface)	Visual reverse-image search for statue identification
Azure Container Apps	Managed revisions	Serverless container hosting with HTTPS ingress
Azure PostgreSQL 16	Standard_B1ms, 32 GB, France Central	Managed relational database
Docker	Desktop + Linux engine	Local build and image push to Docker Hub

Table 6-2 AI Travel Planner technology stack.

Technology	Version / Configuration	Role
FastAPI + Uvicorn	Latest stable	HTTP endpoint definition, request validation, ASGI server
Pydantic	v2.x	PlanRequest, PlaceIn, AgentResponse, TravellerItinerary, and all nested schema validation
CrewAI	Latest stable	Agent and task orchestration for the Events and Planner crews
Groq SDK (via CrewAI LLM)	Latest stable	LLaMA-3.3-70B inference for the Events crew agent
OpenRouter (via CrewAI LLM)	REST API	GPT-4o-mini inference for the Planner crew agent
Tavily Python Client	Latest stable	Web-search tool used by the Events agent to find real upcoming events
requests	Latest stable	HTTP calls to the OSRM routing API (3-second timeout)
math (standard library)	Python stdlib	Haversine-distance fallback calculation when OSRM is unavailable

6.8.2 Key Implementation Details

Statue Storyteller

The Statue Storyteller feature allows a user to upload an image of an Egyptian statue or monument and receive an AI-generated historical narrative in return. The implementation combines computer vision, reverse-image search, large language models, and persistent conversation memory to provide accurate, context-aware responses, drawing on three complementary sources of information when needed: retrieval from the historical knowledge base, a Google Lens reverse-image search, and a general web search.

LangGraph drives every conversational turn. Rather than holding state in memory, the service rebuilds the agent's full context from PostgreSQL on every request, in three stages. First, `load_context()` loads the three most recent messages from the database. Second, this flat list of records is converted into typed `HumanMessage` and `AIMessage` objects and assembled into a `MessagesState` dictionary, with the current user turn appended at the end. Third, the reconstructed state is passed to a LangGraph ReAct agent for inference.

The LangGraph graph is compiled once at startup as a StateGraph with a single ToolNode — wrapping the Google Lens search tool — and a single model node. The model node binds the LLM (served through the Groq SDK) to the tool schema, allowing it to decide mid-turn whether to call Google Lens or answer directly from its own knowledge. When the graph is invoked, it executes the following turn loop: the model node receives the full reconstructed message history and either returns a final text answer or emits a tool-call request; if an image was uploaded, the tool node then performs the visual search and appends the result to the state as a tool message; the model node runs a second pass, producing a final narrative grounded in the identification result; and the graph then ends, at which point the route handler extracts the last assistant message, persists both the user and assistant turns to PostgreSQL, and returns the response to the client.

AI Travel Planner

The AI Travel Planner generates personalized, multi-day itineraries from a user's selected destinations, interests, and available attractions, combining multiple cooperating AI agents with web search and geographic optimization. A user request is first passed to an Events Discovery crew, whose output feeds an Itinerary Planner crew; the resulting draft itinerary is then adjusted using real road travel times from OSRM before the final itinerary is returned as a validated JSON response.

The Planner agent is implemented using the CrewAI framework and an LLM accessed through OpenRouter (GPT-4o-mini). It receives structured input — trip theme, destination city, travel dates, number of days, and a list of candidate attractions with coordinates, estimated visit duration, category, and description — and produces a structured itinerary conforming to a predefined TravelItinerary schema. Its prompt encodes several scheduling rules: geographically close attractions are grouped within the same day to reduce unnecessary travel, the ordering of locations minimizes backtracking based on their coordinates, and approximate travel times are estimated with the Haversine formula assuming an average driving speed of 30 km/h, keeping each day's total schedule to roughly six to nine hours.

The Events agent uses a single CrewAI tool that wraps the Tavily Python client (advanced search depth, two results), configured with delegation and memory disabled for stateless operation. Its prompt directs it to search for concerts, festivals, nightlife, cultural events, and

exhibitions within the trip's date range, with output constrained to a predefined EventsReport JSON schema.

After the itinerary is drafted, a post-processing step invokes OSRM to calculate actual road travel times between consecutive destinations, recalculating the start and end times of subsequent activities for a realistic daily schedule; if OSRM is unavailable or does not respond within the configured timeout, the service automatically falls back to the Haversine-based estimate. Output validity is enforced through CrewAI's structured-output parsing into the TravelItinerary object, with a manual Pydantic validation pass as a fallback whenever automatic parsing fails despite a syntactically valid response, improving robustness against inconsistent model output formatting.

6.9 Data Persistence

Each module owns and migrates its own database independently through Entity Framework Core migrations, following a database-per-module boundary that mirrors the module boundaries in the code itself. Mapping and constraint concerns are kept out of the domain entities and placed instead in dedicated EntityConfig classes, so that the domain model remains focused purely on business behavior.

6.10 Web Administration Dashboard

The administration dashboard used by TouRA staff to manage users, companies, places, support requests, and upgrade requests is a single-page application built with React and TypeScript and bundled with Vite. The codebase is organized by feature area (for example, usersManagement, companiesManagement, placesManagement, supportManagement, upgradeRequests, and trips), with Redux Toolkit and RTK Query used to fetch, cache, and synchronize server data for each of these areas, and Leaflet used to render interactive maps for reviewing and picking place locations.

6.11 Mobile Application

The tourist-facing mobile application is built with React Native and Expo, using Expo Router for file-based, group-oriented navigation across the authentication, onboarding, traveler, and guide experiences. Server data is managed with React Query, while lightweight local and UI state (such as the in-progress trip-creation wizard and authentication state) is managed with Zustand stores. To support unreliable connectivity in the field, the app maintains a local SQLite database that mirrors conversations, messages, and pending outgoing messages, together with an offline queue processor that automatically re-sends unsent messages once the connection is restored. Real-time chat and notifications are received through dedicated SignalR client connections, and a set of Jest unit tests cover key utilities and components — including trip-card rendering, the trip-creation store, and environment/formatting helpers — providing an additional layer of automated verification alongside the manual, front-end-driven testing described in Chapter 6.

6.12 Tools and Technologies

Table 5-3 summarizes the main tools and technologies used across the backend, the AI microservices, the administration dashboard, and the mobile application.

Table 6-3 Tech Stack for the whole App

Layer / Concern	Technology	Reason for Selection
Backend framework	ASP.NET Core (.NET), organized per module using Clean Architecture	Strong typing, mature tooling, and clear separation of Domain, Application, Infrastructure, and Presentation layers, which keeps each module independently maintainable.
Data access	Entity Framework Core, one DbContext per module	Code-first migrations and LINQ-based querying, with a database-per-module boundary that keeps each service's data model independent.
Identity & access management	Custom multi-role authentication service issuing JWT access tokens	Keeps registration, login, OTP verification, and role-based authorization fully in-house, avoiding the operational overhead of an external identity provider for a project with a small, well-defined set of roles.
Inter-module messaging	Transactional Outbox / Inbox pattern over a shared event bus	Guarantees at-least-once, idempotent delivery of integration events between modules even if the broker or a consumer is briefly unavailable.
Object storage	MinIO (S3-compatible object storage)	Self-hostable and API-compatible with common cloud storage, used to store user-uploaded images and documents outside the relational databases.
Real-time communication	SignalR (ChatHub, NotificationHub, TripHub)	Native ASP.NET Core integration for WebSocket-based push updates, avoiding client-side polling for chat, notifications, and live guide tracking.
Containerization	Docker and Docker Compose	Consistent local development and deployment environments across the multiple services and supporting infrastructure (databases, message broker, and object storage).
Admin dashboard	React, TypeScript, Vite, Redux Toolkit with RTK Query, Leaflet	Fast development experience, predictable and cache-aware server-state management, and mature mapping components for location-based screens.
Mobile application	React Native with Expo, Expo Router, React Query, Zustand, expo-sqlite	Cross-platform delivery from a single codebase, file-based navigation, offline-first local persistence, and lightweight state management suited to a mobile client.
AI / ML integration	DINOv2 vision encoding, Google Lens reverse-image search, Groq (LLaMA) and OpenRouter (GPT-4o-mini) via LangGraph and	Combines visual feature extraction, fast open-model inference, and multi-agent orchestration to power monument recognition and itinerary generation; detailed further in this chapter.

Layer / Concern	Technology	Reason for Selection
	CrewAI, Tavily search, OSRM routing	

7 TouRA Testing and Evaluation

This chapter presents the testing approach adopted for TouRA, the executed test cases and their results, and an evaluation of the project in terms of cost, time, environmental impact, and social and political impact.

7.1 Testing Methodology

Given the project's modular, multi-service architecture, testing was carried out primarily as manual, end-to-end verification driven from the front end, rather than through isolated unit tests written against each service in the back end. For every functional requirement and use case identified in Chapter 3, the corresponding user journey was executed directly through the mobile application: creating an account, generating a plan, requesting a guide, uploading a monument photo, and so on. Each flow was performed as a real user would perform it, and the outcome was verified by observing the resulting UI state and confirming that the flow completed correctly end to end across the modules it touched.

This front-end-first approach was chosen because the most meaningful indicator of correctness for TouRA is whether a complete user flow behaves as expected once all the modules involved — Users, Trips, Conversations, Notifications, and Payments — are exercised together through the real client, rather than in isolation. Where a flow surfaced unexpected behavior, Postman and direct database inspection were used to confirm the specific request or record involved.

Testing covered the happy path for each use case as well as the alternate flows documented in the use case tables of Chapter 3, such as invalid credentials, unrecognized monuments, and rejected bids. Table 6-1 summarizes the main test cases executed and their outcomes.

7.2 Test Cases and Results

Table 7-1 Summary of executed test cases and results.

Test ID	Related Requirement	Front-End Flow Tested	Expected Result	Actual Result	Status
TC-1	UC-1 Register Account	User opens the app, taps “Sign Up”, enters email/phone and password, receives and enters the OTP.	Account is created, OTP is verified, and the user is redirected to profile setup.	Account created and OTP verified successfully; user redirected to profile setup as expected.	Pass
TC-2	UC-2 Login	User enters registered credentials on the login screen and submits.	User is authenticated and redirected to home, or to profile setup if incomplete.	Login succeeded; redirection logic matched the profile-completion flag returned by the User Service.	Pass
TC-3	Forgot Password / Reset	User requests password reset, receives OTP by email/SMS, verifies it, then sets a new password.	Password is updated and the reset token is invalidated after use.	New password accepted; reusing the same OTP/token was correctly rejected on a second attempt.	Pass
TC-4	UC-3 Generate AI Trip Plan	User selects a theme (e.g., romantic, family), sets location and budget, and requests an AI-generated plan.	A plan with correctly themed places is generated and shown on the interactive map.	Plan was generated and displayed with places matching the selected theme’s category distribution.	Pass
TC-5	UC-4 Join a Generated Plan	User browses public plans, opens one with available capacity, and taps “Join”.	User is added as a participant and gains access to the plan’s group chat.	User was added as a participant; group chat became accessible immediately after joining.	Pass
TC-6	UC-5 Guide Bidding and Assignment	Plan owner requests a guide; guide submits a bid; owner reviews bids and accepts one.	The accepted guide is assigned, other bids are auto-rejected, and the plan is locked.	Selected guide was assigned and all participants were notified; remaining bids were rejected automatically.	Pass
TC-7	UC-6 Landmark Recognition	User photographs a monument in the app and sends the image to the AI assistant.	The system identifies the monument and returns historical/cultural information in the chat.	Recognized monuments returned accurate information; unrecognized images triggered the fallback retry prompt.	Pass
TC-8	UC-7 Place Rating	User selects a visited place from a completed plan and submits a rating with a comment.	The review is stored and the place’s average rating is updated.	Review was saved and reflected in the place’s rating on the next screen load.	Pass

Test ID	Related Requirement	Front-End Flow Tested	Expected Result	Actual Result	Status
TC-9	UC-8 Create Featured Plan	A verified guide or company fills in the plan-creation form with places and submits it.	The plan is validated and published as a featured plan.	Plan was published successfully; submitting an incomplete form correctly returned a validation error.	Pass
TC-10	Real-Time Chat and Notifications	User sends a message in a trip/AI chat and triggers an action that generates a notification (e.g., bid accepted).	Messages and notifications are delivered in real time via the SignalR hubs.	Messages and notifications arrived instantly on connected clients; reconnection after a dropped connection restored the session automatically.	Pass

All executed test cases produced the expected outcome. Where an alternate flow was tested (for example, submitting an already-registered email, or photographing an unrecognized artifact), the system responded with the appropriate error message or fallback behavior defined in the corresponding use case. No blocking defects were identified during this round of testing; minor UI inconsistencies observed during testing were logged and corrected before the next iteration, consistent with the Agile methodology adopted for the project.

7.3 Project Evaluation

Beyond functional correctness, the project was evaluated against several broader criteria to assess its overall value and feasibility.

7.3.1 Cost

As a graduation project, TouRA was developed without a dedicated financial budget; all development, design, and testing effort was contributed by the team members. The main recurring costs identified for a production deployment would be cloud hosting for the modules and the AI microservices, managed database and container hosting on Azure, object storage for images, and usage-based fees for the LLM providers (Groq and OpenRouter) used for AI chat, itinerary planning, and monument recognition. These costs scale with active users and API calls rather than being fixed, which makes the platform economically feasible to pilot at a small scale before committing to larger infrastructure spending.

7.3.2 Time Compared to Contemporary Systems

Under the current fragmented approach, a tourist typically needs a route-planning application, a review platform, a separate booking channel, and an independently arranged local guide, each requiring its own search and setup time. During informal comparison, arranging an equivalent themed outing using three to four separate applications took noticeably longer than generating and adjusting a themed plan directly within TouRA, since the platform consolidates location discovery, itinerary generation, guide bidding, and booking into a single flow. This reduction in planning time was one of the core motivations behind the project, as reflected in the survey results discussed in Chapter 2.

7.3.3 Environmental Impact

By consolidating trip planning into a single digital platform, TouRA reduces reliance on printed maps, guidebooks, and paper tickets. Theme-based plan generation also encourages tourists to group nearby activities together, which can reduce unnecessary transportation between poorly sequenced locations. These effects are modest at the scale of a graduation project but would compound with wider adoption.

7.3.4 Social and Political Impact

TouRA supports Egypt's broader tourism development goals by making cultural heritage more accessible and by giving independent, verified guides a structured marketplace in which to offer their services alongside established companies. The rating and verification mechanisms aim to reduce the transportation and guiding scams frequently reported by tourists, which can improve visitor trust and, over time, contribute positively to Egypt's international tourism reputation. By supporting both foreign visitors and Egyptian citizens, the platform also promotes wider domestic engagement with national heritage sites.

8 Conclusions and Future Work

8.1 Conclusion

This project set out to address a clear gap in the tourism technology market: the absence of a single, intelligent platform that unifies theme-based trip planning, trusted guide coordination, and interactive cultural exploration for tourists visiting Egypt. TouRA was designed and implemented as an AI-powered smart tourism platform that combines a theme-based trip planner, a guide bidding marketplace, real-time map visualization, a ratings and review system, and AI-driven features for monument recognition and conversational trip planning, all built on a modular architecture covering user management, trips, conversations, notifications, and payments.

Testing, described in Chapter 6, confirmed that the core user journeys defined in Chapter 3 — registration and authentication, AI plan generation, joining a plan, guide bidding, landmark recognition, place rating, and featured plan creation — function correctly end to end when driven through the actual mobile application. This validates that the system, in its current state, meets the primary objectives set out in Chapter 1: it allows users to generate and join theme-based plans, connects them with bidding guides, visualizes trips on an interactive map, and integrates AI capabilities for both monument recognition and conversational plan requests.

The project also surfaced the practical challenges anticipated in Chapter 1, particularly around the absence of ready-made datasets for Egyptian artifacts and trusted tourism service providers. Building a curated image dataset of 75 monuments and implementing the Statue Storyteller and AI Travel Planner microservices — combining DINOv2-based visual retrieval, Google Lens reverse-image search, and LLM reasoning orchestrated through LangGraph and CrewAI — required considerably more effort than a typical classification or chatbot task, but ultimately produced a system capable of handling the open-ended, culturally specific questions tourists are likely to ask.

Overall, the graduation project demonstrates that an integrated, AI-driven approach to tourism planning and guidance is technically feasible and directly responsive to the pain points

identified both in the literature survey and in the team's own user survey, and it lays a working foundation that can be extended toward a production-ready product.

8.2 Data and Budget Limitations

Several of the project's limitations stem directly from the fact that it was developed without external funding or institutional data-sharing agreements, and it is worth stating these constraints explicitly.

On the data side, the 75-monument image dataset described in Chapter 4 was built entirely through manual, unpaid effort: images were gathered from whatever public sources the team could find and individually verified for historical accuracy, since there was no budget to commission original museum photography, license existing archival collections, or purchase a ready-made artifact dataset. This constrained both the number of monuments covered and the number of images available per artifact, which in turn limited how much the recognition model could be trained and validated beyond the roughly 15-images-per-class minimum discussed earlier. A similar issue affected the trusted-services side of the platform: without funding to acquire or verify a large hospitality and guiding directory, the team could not build the comprehensive, vetted database of hotels, restaurants, and tour operators that the market analysis in Chapter 2 identified as missing from existing solutions.

The lack of budget also shaped several concrete technical decisions in the AI microservices described in Chapter 5. The Statue Storyteller relies on an unofficial Google Lens interface for reverse-image search rather than a licensed visual-search API, and the AI Travel Planner relies on the free, open-source OSRM routing engine rather than a paid mapping and directions provider; both choices kept the project viable without a hosting or API budget, but at the cost of the reliability guarantees, rate limits, and support that a paid service would normally provide. Similarly, LLM inference was built around Groq and OpenRouter specifically because they offered usable free or low-cost tiers, rather than because they were necessarily the strongest models available. As a result, some of the performance targets defined in Chapter 3 — such as AI response time and monument-recognition accuracy — were validated informally during manual testing rather than under sustained, production-level load, since doing so would itself require paid infrastructure the project did not have access to.

These constraints do not undermine the validity of the system design or the testing results presented in Chapter 6, but they do mean that the current dataset, third-party service tiers, and infrastructure should be treated as a proof-of-concept foundation rather than a production-ready configuration, and several of the future-work items below are aimed directly at addressing them.

8.3 Future Work

While the current version of TouRA satisfies its core objectives, several directions were identified for future development beyond the scope of this graduation project:

- **Data partnerships:** Pursue partnerships with the Ministry of Tourism and Antiquities, museum authorities, or tourism boards to gain access to larger, higher-quality artifact image collections and verified service-provider directories, directly addressing the data limitations described above.
- **Dataset expansion:** Extend the artifact recognition dataset beyond the current 75 monuments to cover additional museums, archaeological sites, and regional heritage locations across Egypt, improving recognition coverage and accuracy.
- **Production-grade infrastructure:** Move from the free-tier OSRM routing engine, unofficial Google Lens interface, and Groq/OpenRouter LLM usage to paid, production-grade routing, vision, and language-model subscriptions once funding allows, to properly validate and meet the latency and accuracy targets defined in Chapter 3 under real load.
- **Production payment integration:** Replace the current payment abstraction with a fully integrated local payment gateway to support real transactions for bookings, guide fees, and featured plans.
- **Trusted service provider database:** Build a verified, continuously maintained database of hotels, restaurants, and tour operators, rather than relying solely on user-submitted reviews, to further improve recommendation quality and trust.
- **Offline support:** Add offline access to previously generated plans and downloaded monument information for use in areas with limited connectivity, which is common at remote archaeological sites.

- Expanded language support: Extend multilingual support beyond the languages currently handled by the conversational AI to serve a wider international tourist base.
- Authentication and role management enhancements: Extend the existing multi-role JWT system with refresh-token rotation, finer-grained permissions, and an administrative role-management interface as the platform's user base and role types grow.
- Guide verification enhancements: Introduce AI-assisted document verification for guide onboarding to streamline and strengthen the trust and safety mechanisms of the guide marketplace.
- Scalability testing: Conduct formal load and performance testing against the non-functional targets defined in Chapter 3 (2,000 concurrent users, sub-two-second map rendering, five-to-eight-second AI response times) as the user base grows and production-grade infrastructure becomes available.

Together, these enhancements would move TouRA from a validated graduation project prototype toward a production-grade smart tourism platform capable of serving a wider audience of tourists and tourism service providers in Egypt.